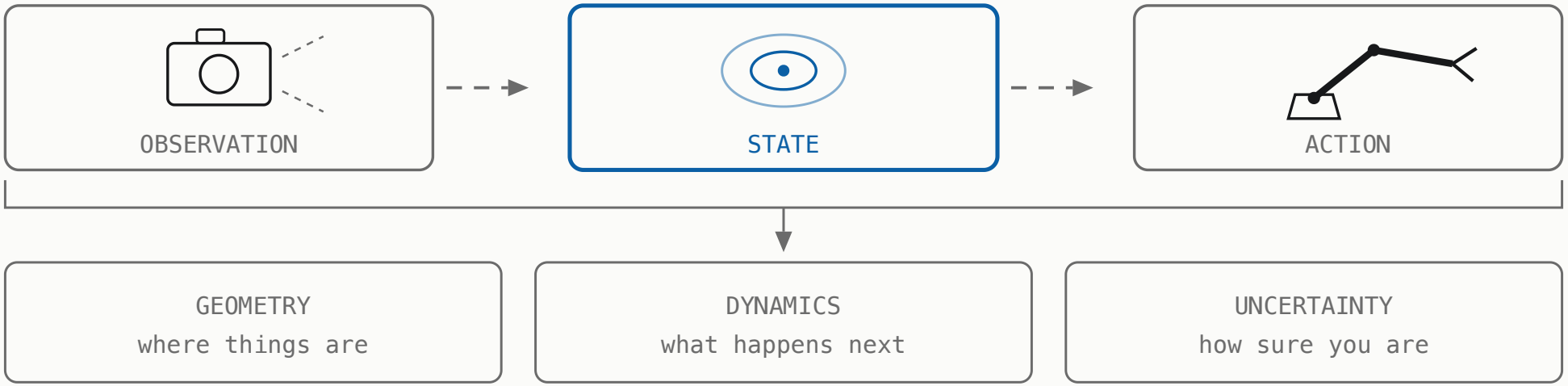
# Introduction and Linear Algebra I

## Spaces, Maps, Rank, Solutions

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon

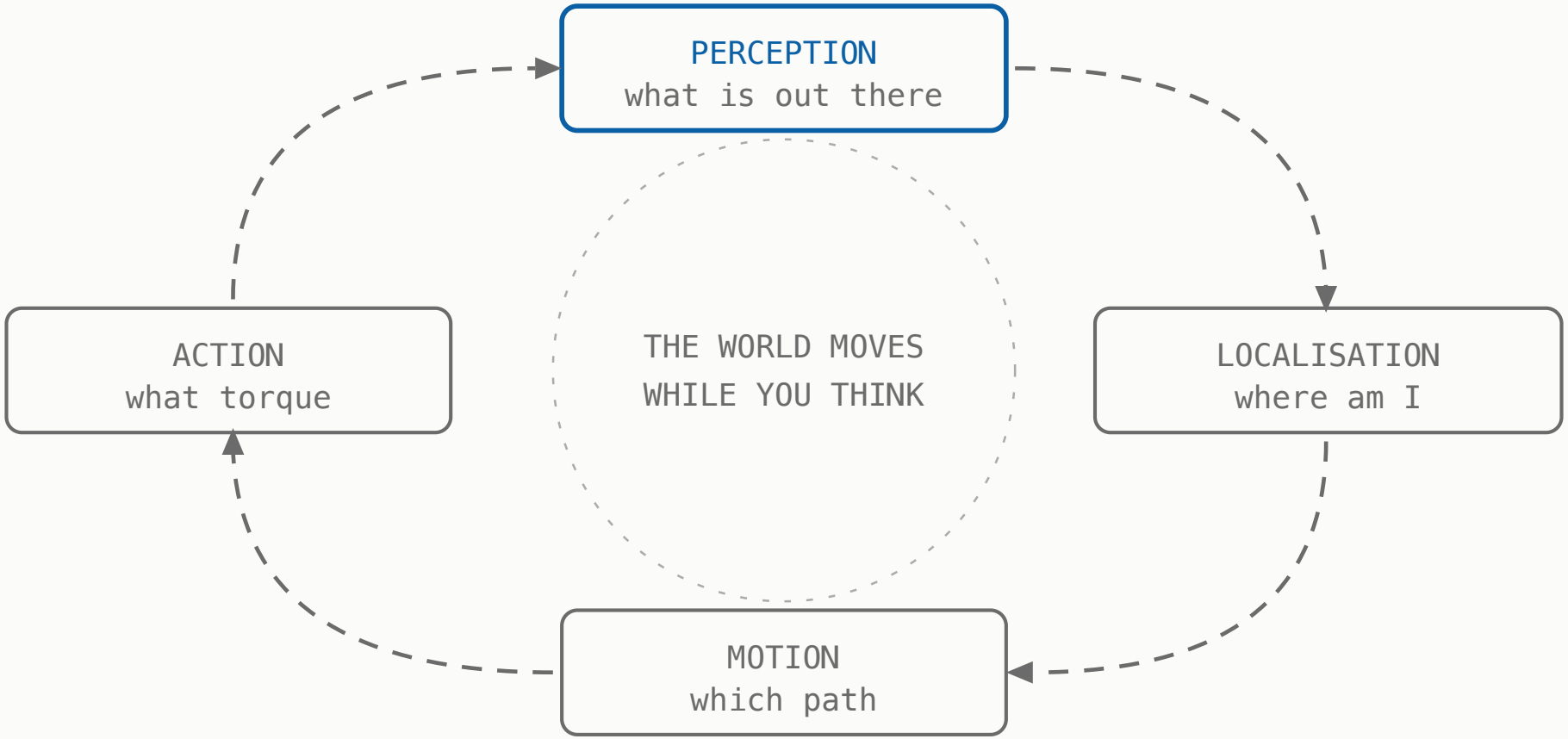
Jing Zhang · Fall 2026 · September 2

# Why Mathematics for Robotics?



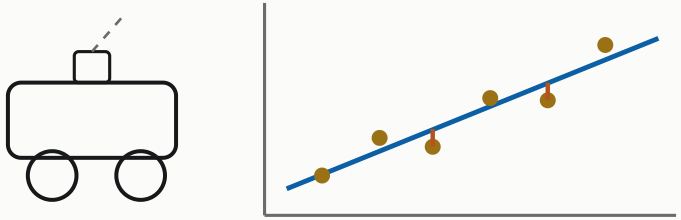

The three panels are recurring questions, not one per box above.

# Why Mathematics for Robotics?



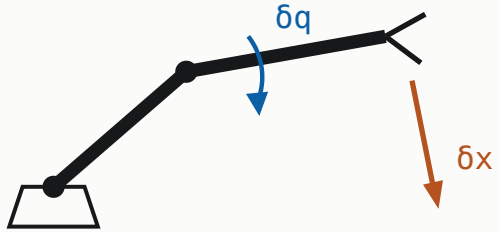

The loop closes many times a second.

# Why Mathematics for Robotics?



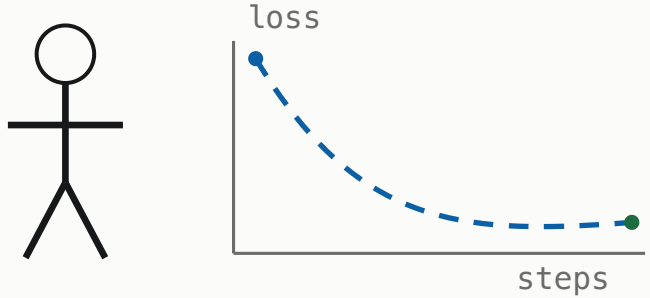
CALIBRATION

locally, $Ax \approx b$




ROBOT MOTION

$\delta x \approx J\,\delta q$




LEARNING

$\arg\min_\theta \mathcal{L}(\theta)$

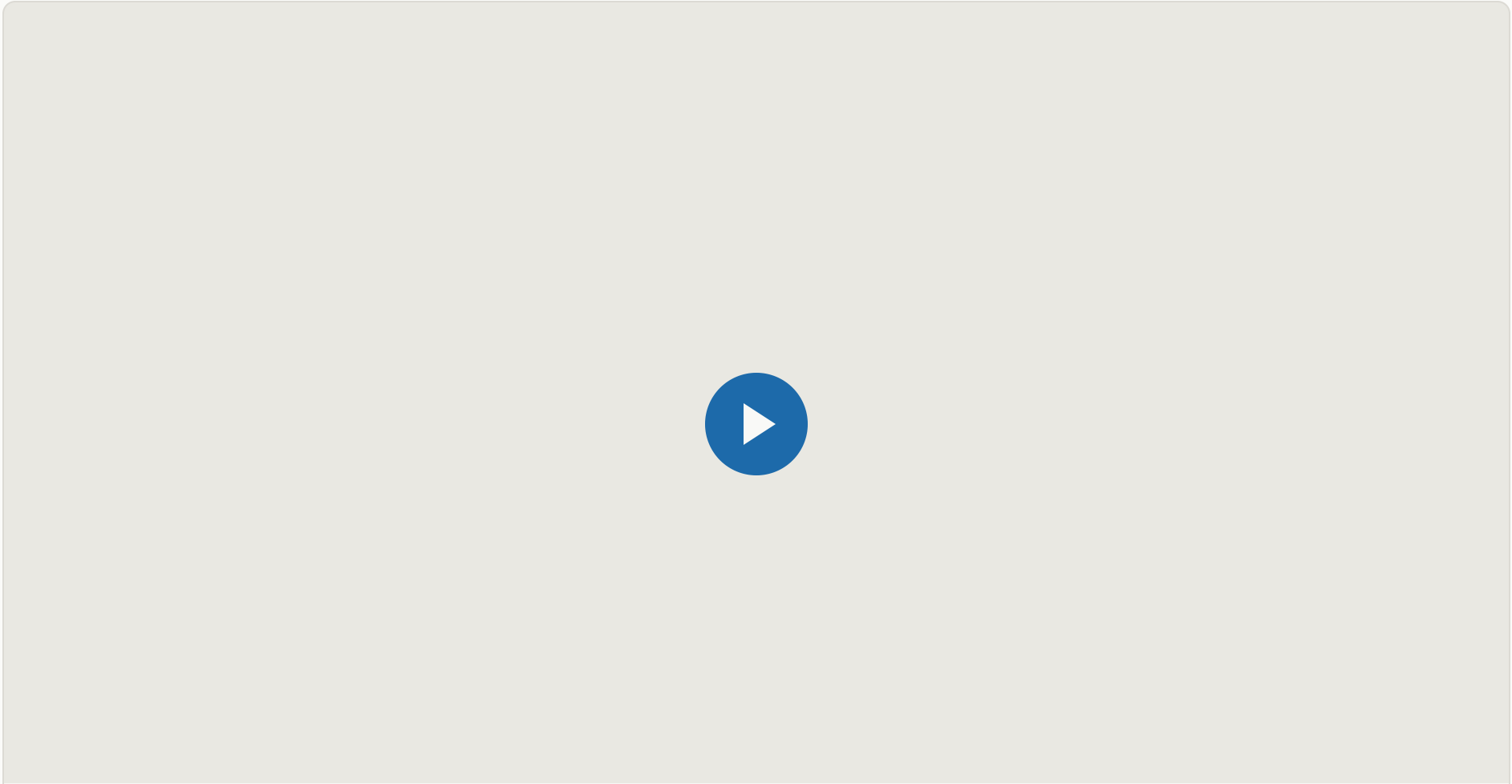
> Different surface, same mathematical structure.

# Why Mathematics for Robotics?

Robotics changes quickly.

The mathematical structure changes much more slowly.

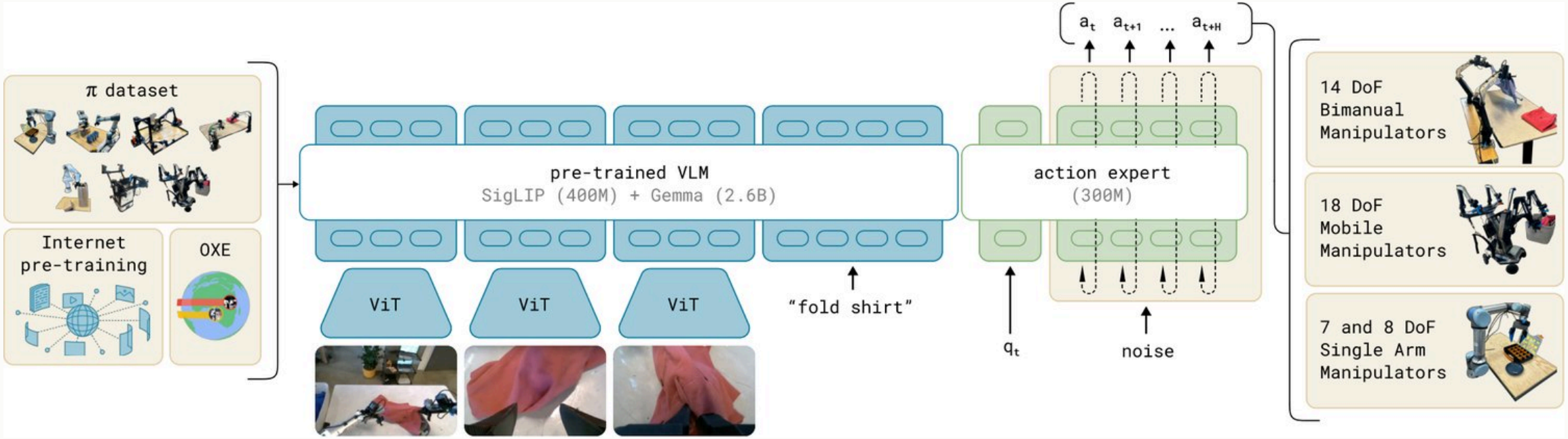
# Why Mathematics in the GenAI Era?



Google DeepMind · Advanced dexterity with Gemini Robotics 2, July 2026

opens on YouTube

# Why Mathematics in the GenAI Era?

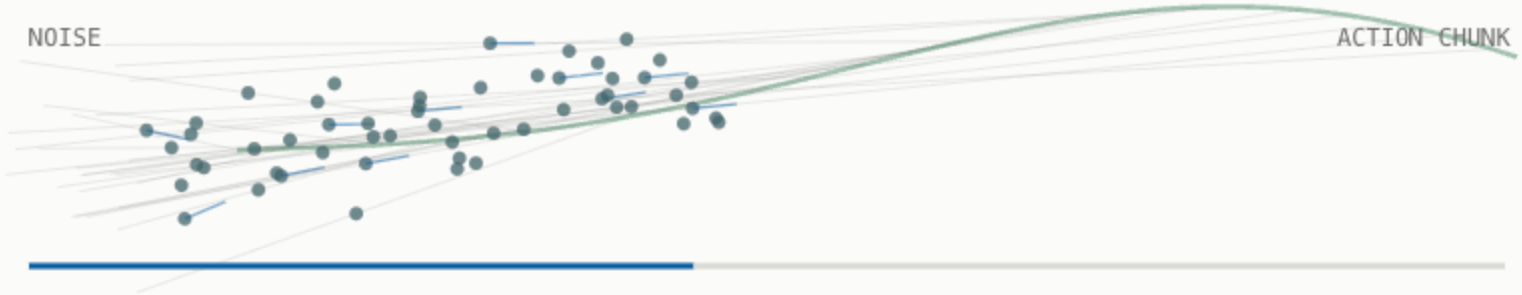



**VLM** · attention, masks · Transformers  $\mathbf{q_t}$ · vector spaces · Linear Algebra I–IV  **noise** → $\mathbf{a_t}$ · a vector field · ODE · Flow Matching

A modern VLA is not replacing mathematics. It is built from mathematics.

Black, K. et al. ***π0: A Vision-Language-Action Flow Model for General Robot Control***, Figure 3. RSS, 2025. arXiv:2410.24164.

# Why Mathematics in the GenAI Era?

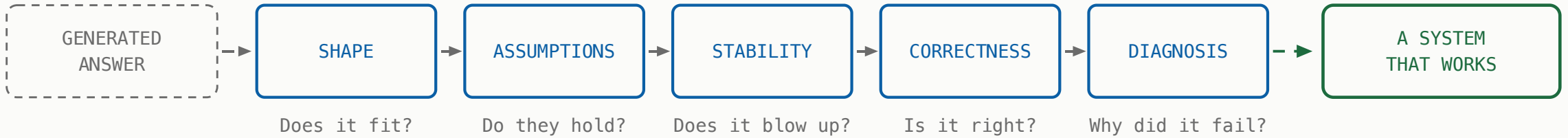



$$\boxed{\frac{dA_\tau}{d\tau} = v_\theta(A_\tau, o, \tau)} \qquad A_0 = \epsilon, \qquad \hat{a} = A_1$$

- $\tau$ flow time, 0 to 1; the field changes with it
- $v_\theta$ the trained vector field
- $A_0$ the start: pure noise $\varepsilon$
- $\hat{a} = A_1$ the action chunk, read off at $\tau = 1$

Black, K. et al. ***π0***. RSS, 2025. arXiv:2410.24164.

# Why Mathematics in the GenAI Era?




> AI can generate. You still have to judge.

# Why Mathematics in the GenAI Era?

| WHAT AN ML INTERVIEW ASKS ABOUT | | |
|---|---|---|
| **Rank / span** ● | **Null / image** ● | **Independence** ● |
| **Inverse** ● | **Singular** ● | Orthogonality |
| Positive semidefinite | Eigenvalues | Jacobian |
| Hessian | Determinant | Dot product |

| AND TO WRITE, BY HAND | | Lectures 5 · 6 · 10 |
|---|---|---|
| transformer end-to-end | causal, cross, self attention | flash attention |
| attention backward pass | MLP forward and backward | SGD training loop |

> "...with AI assistance completely off."

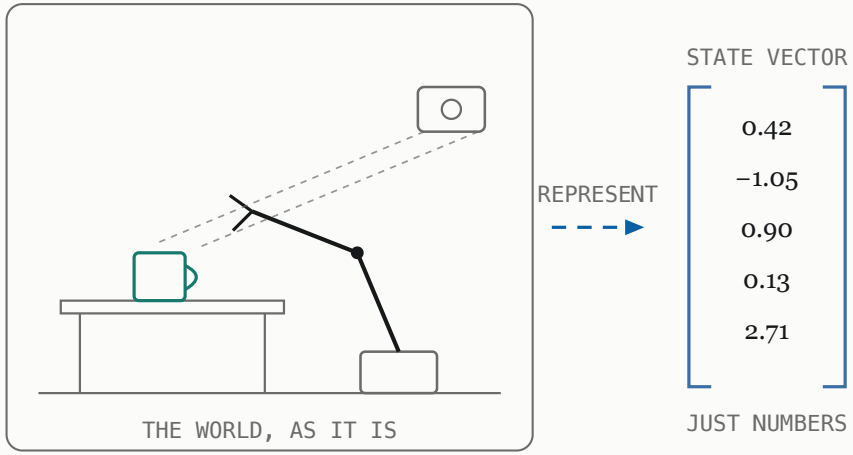
Sapora, S. ***ML Job Interviews: The Ultimate Guide***, 2026 · silviasapora.github.io/blog/ml-interviews.html
Liu, A. ***Notes on the Industry Job Search***, 2026 · alisawuffles.github.io/blog/job-search/

# The Math We Keep Coming Back To

| | | |
|---|---|---|
| $x$ | state | Linear Algebra I · ODE · Probability I |
| $A$ | linear map | Linear Algebra I–IV |
| $f$ | nonlinear function | ML Math I · ODE |
| $J$ | Jacobian | Linear Algebra IV · ML Math I–II |
| $p$ | probability distribution | Probability I–II · Diffusion |
| $L$ | loss | ML Math II · Diffusion · Flow Matching |

> Different problems, the same mathematical ideas.

# The Math We Keep Coming Back To




**FOUR ROBOT QUANTITIES**

| | |
|---|---|
| $q \in \mathbb{R}^7$ | joint configuration |
| $p \in \mathbb{R}^3$ | position |
| $I \in \mathbb{R}^{3\times H\times W}$ | image |
| $a \in \mathbb{R}^{16\times 7}$ | action chunk |

Each is a vector, or an array that can be vectorized.

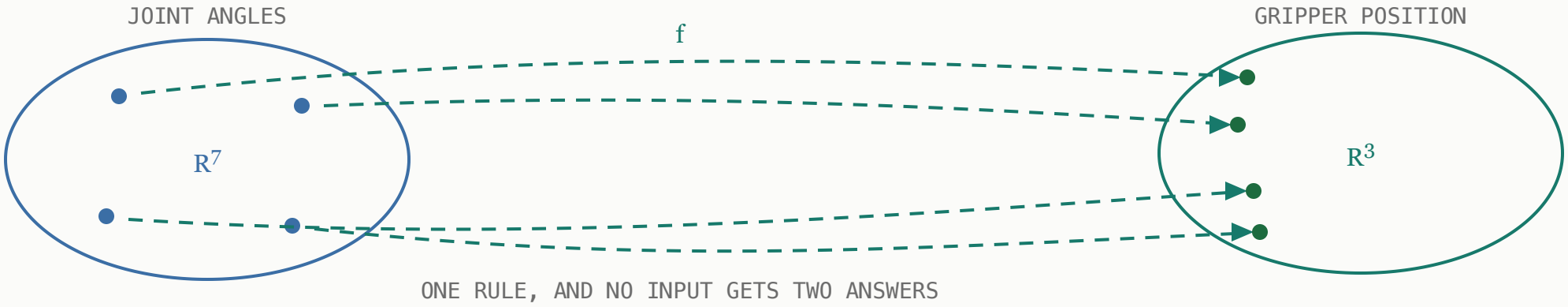
# The Math We Keep Coming Back To

MAP $f : \mathbb{R}^n \to \mathbb{R}^m, \qquad u \mapsto y = f(u)$

`domain` $\mathbb{R}^n$, the allowed inputs

`codomain` $\mathbb{R}^m$, the space the outputs live in

`u ↦ y` the rule that carries one to the other




Forward kinematics: $\mathbb{R}^7 \to \mathbb{R}^3$, joint angles to gripper position.

# The Math We Keep Coming Back To

LOCAL LINEARISATION

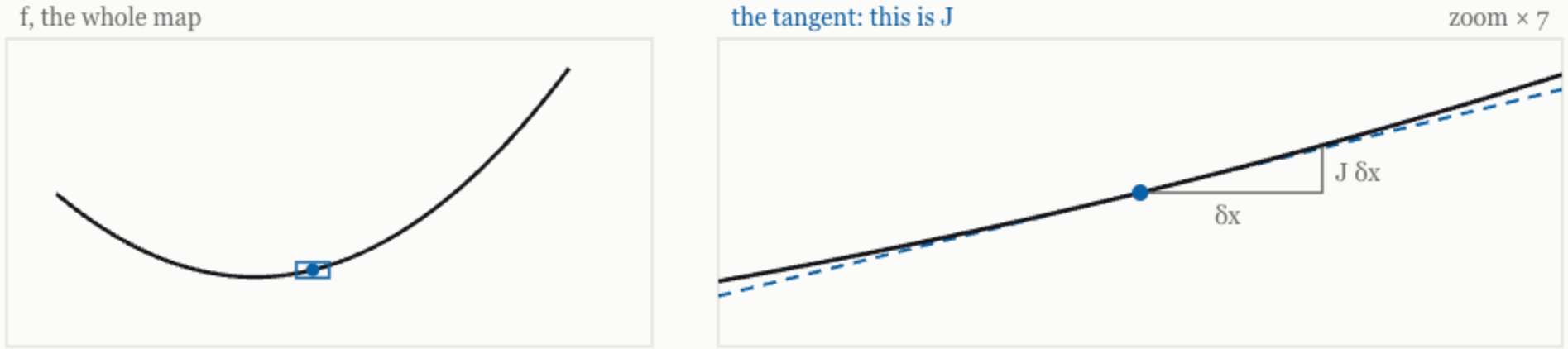
$$f(x + \delta x) \approx f(x) + J(x)\,\delta x$$

`x` the current point

`δx` a small change

`J(x)` the Jacobian, $J(x) \in \mathbb{R}^{m \times n}$

> Many nonlinear robotics problems become linear when viewed locally.

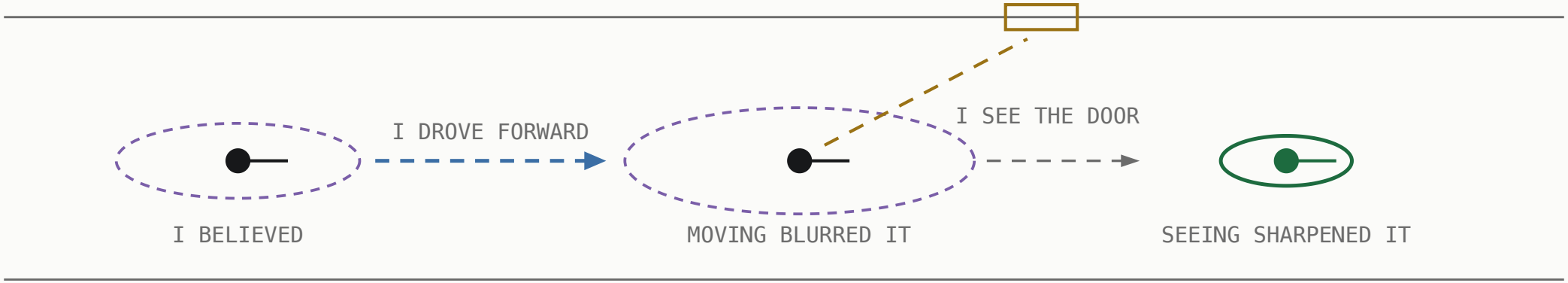



Left, the shrinking window. Right, what is inside it.

# The Math We Keep Coming Back To

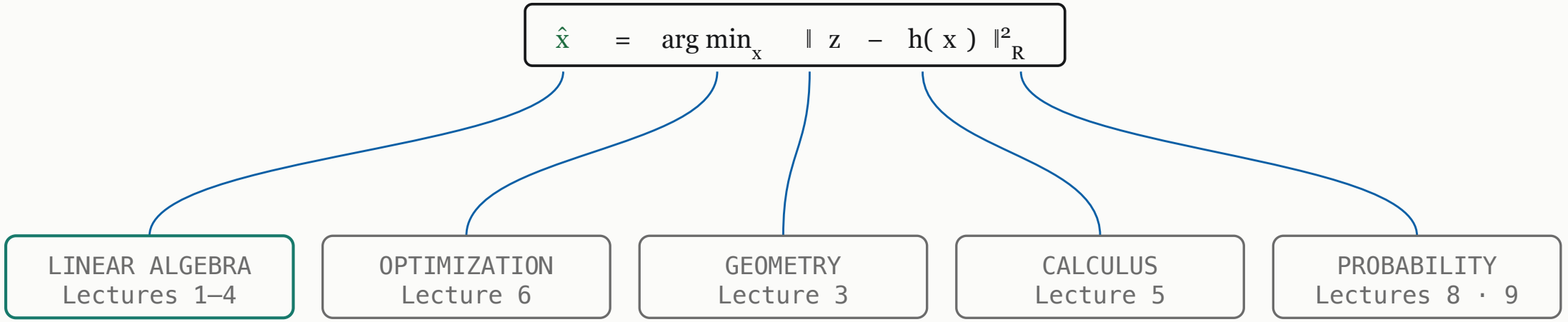
WHERE AM I? $x_t = f(x_{t-1}, u_t) + w_t, \qquad z_t = h(x_t) + v_t$

- $x_t$ the latent state, usually not observed directly
- $u_t$, $z_t$ the command, the measurement
- $f$, $h$ motion and sensor models
- $w_t$, $v_t$ noise




Thrun, S., Burgard, W., Fox, D. ***Probabilistic Robotics***, Ch. 7. MIT Press, 2005.

# The Math We Keep Coming Back To




$\|r\|^2_R = r^\mathsf{T} R^{-1} r$, with $R$ the measurement covariance.

r $z - h(x)$: measurement minus prediction; $R$ is positive definite

Barfoot, T. D. ***State Estimation for Robotics***, Chapter 4. Cambridge University Press, 2017.

# Course information

THE MEETING

| | |
|---|---|
| WHEN | Wednesdays 5:00–7:30 PM |
| WHERE | 2 MetroTech Center, Room 909 |
| ONLINE | nyu.zoom.us/j/95222620311 |
| UPDATES | slides and recordings on the course page |
| QUESTIONS | Ed Discussion, linked from the course page |

THE STAFF

**Jing Zhang**
instructor · z.jing@nyu.edu

**Shichen Zhang**
teaching assistant · sz4968@nyu.edu

office hours to be announced

Everything on these four pages, plus the schedule and the reading: jingz6676.github.io/rob6013.html

# Course information

GRADING




FOUR PROBLEM SETS
understanding, not recall

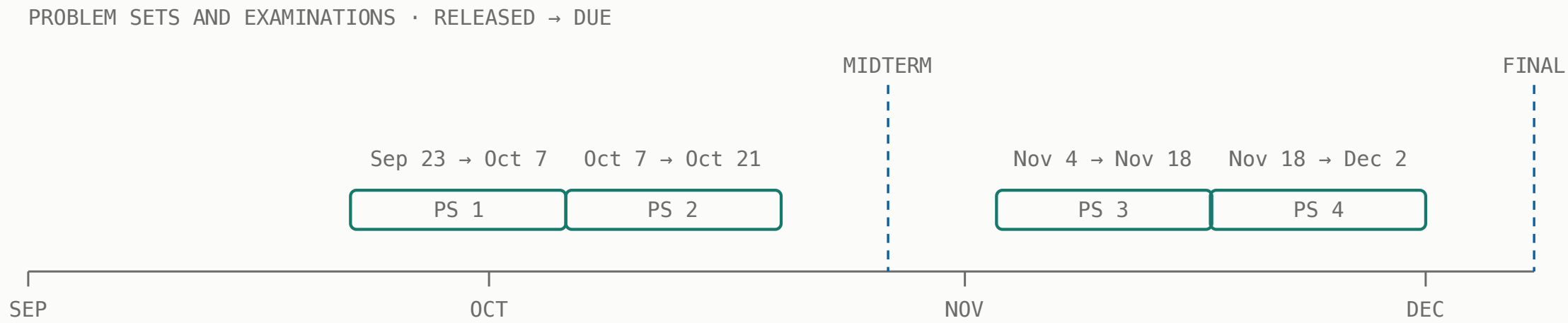
MIDTERM
28 October, in person

FINAL
9 December, in person

---

up to +10% on the course grade, for sets typeset in LaTeX or Markdown

# Course information




> One sheet of paper, both sides, prepared by you, and a scientific calculator. No phone, laptop, tablet, or watch.

# Course information

| POLICIES | |
|---|---|
| LATE WORK | 10% of the score per calendar day; closed once solutions are posted |
| COLLABORATION | discuss with anyone, then write your solution alone |
| **USING AI** | any tool to understand it, none to write your solutions |
| ACCOMMODATIONS | through the Moses Center, with no need to tell me why |
| IF IT GOES WRONG | email me and Shichen |

LINEAR ALGEBRA I

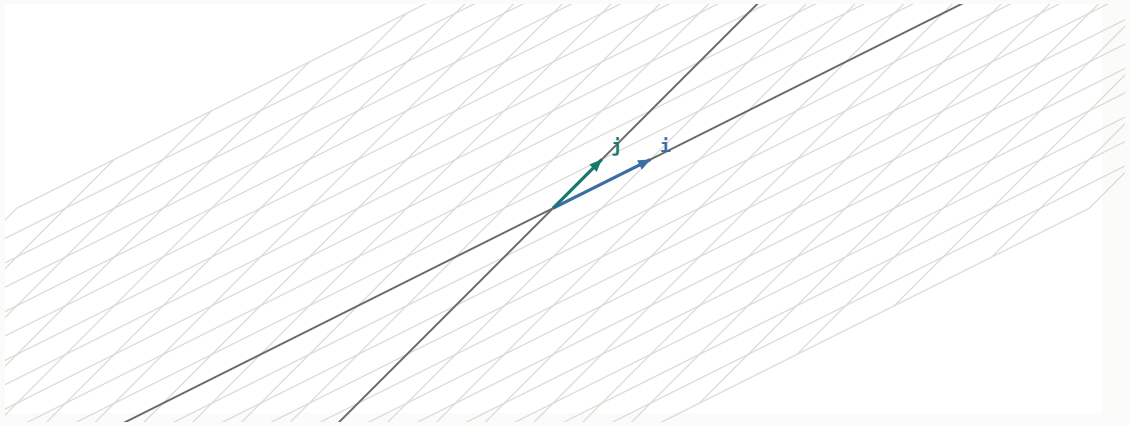
# Spaces, Maps, Rank, Solutions

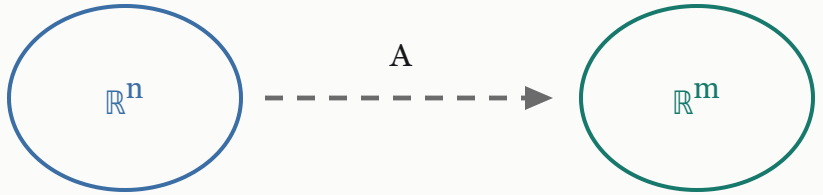


# A matrix is a map

$$A \in \mathbb{R}^{m \times n}, \qquad x \in \mathbb{R}^n \longmapsto Ax \in \mathbb{R}^m$$

- `A` the map itself, m rows by n columns
- `x` the input, a column of n numbers
- `Ax` the output, a column of m numbers
- `n` input dimension: the number of columns
- `m` output dimension: the number of rows; m and n need not be equal



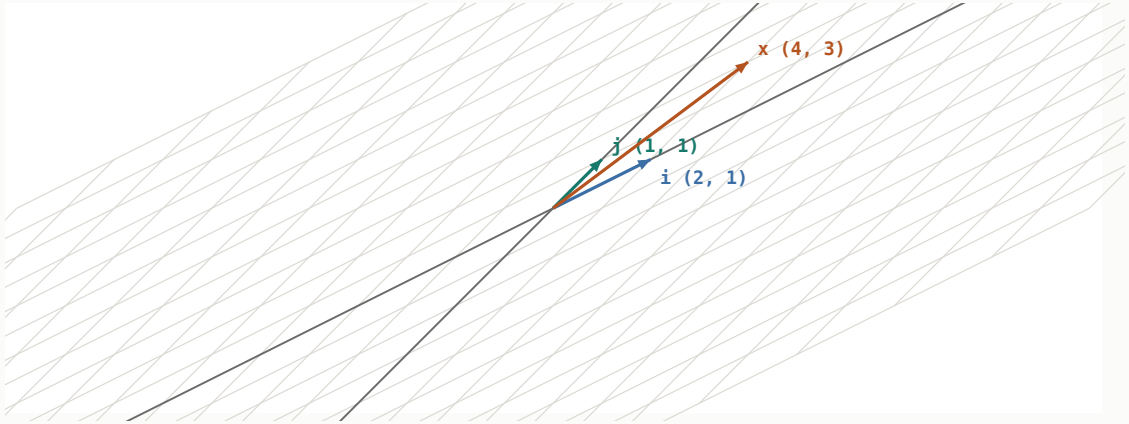

Sanderson, G. ***Essence of Linear Algebra***, ch. 3. 3Blue1Brown, 2016.

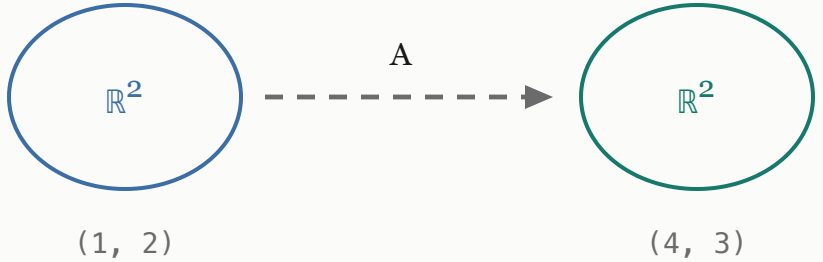
# A matrix is a map

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+2 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ the first column, where $e_1$ lands

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ the second column, where $e_2$ lands



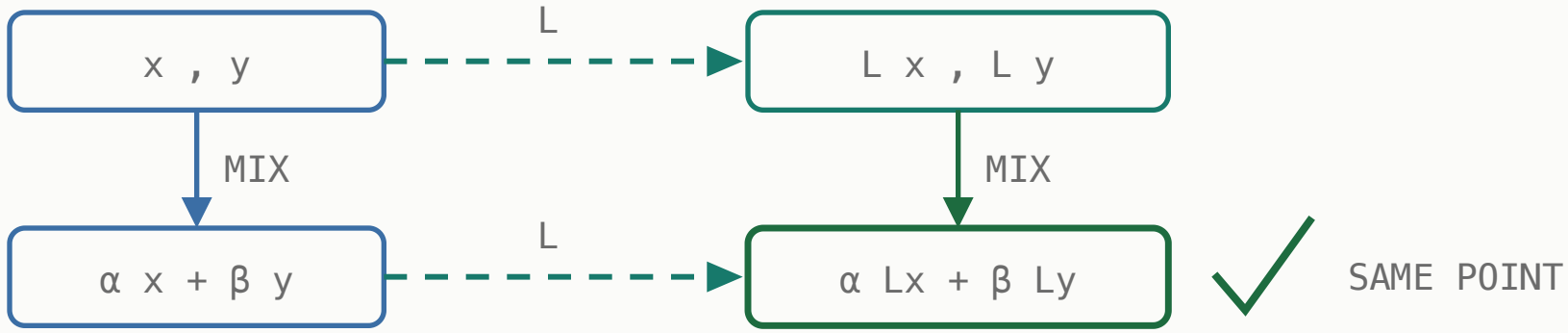
# A matrix is a map

**LINEAR MAP** $\boxed{L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)}$

**additivity** $L(x + y) = L(x) + L(y)$, the case $\alpha = \beta = 1$

**homogeneity** $L(\alpha x) = \alpha L(x)$, the case $\beta = 0$

**superposition** both at once, for all $x, y$ and all $\alpha, \beta \in \mathbb{R}$

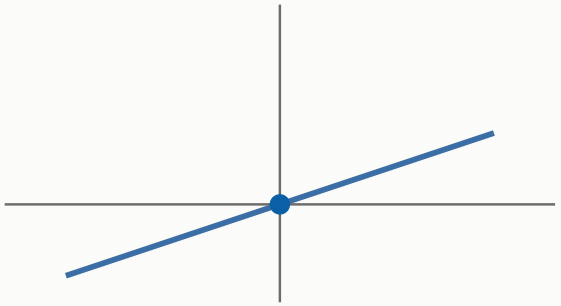



Axler, S. ***Linear Algebra Done Right***, 4th ed., Definition 3.1, p. 52. Springer, 2024. Open access.

QUESTIONS

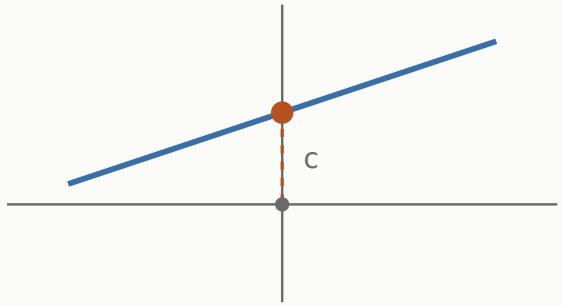
# Which of these three maps is linear?

$f(x) = a\,x$



LINEAR

$f(x) = a\,x + c, \quad c \neq 0$

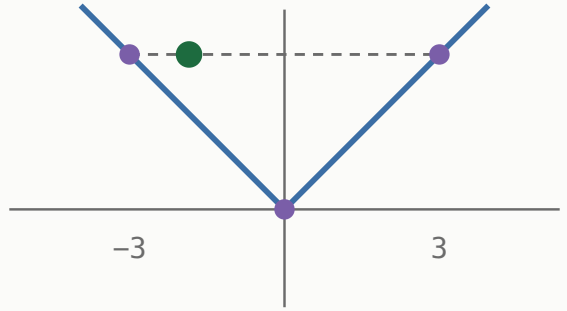



AFFINE, NOT LINEAR

$a = 1, \quad c = 2$

✗ $f(0) = 2 \neq 0$

$f(x) = |x|$




NOT LINEAR

✓ $f(0) = 0$

✗ $f(-3) = 3, \quad -f(3) = -3$

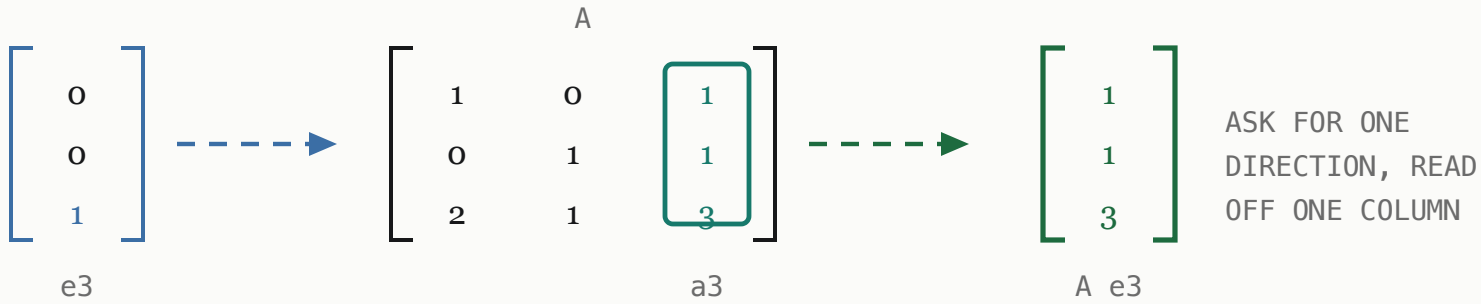
# A matrix is a map

$$e_j = (0, \ldots, 1, \ldots, 0)^{\mathsf{T}}$$

`the 1` sits in slot j; every other slot holds 0

COLUMN RULE $\boxed{Ae_j = a_j}$

$a_j$ column j of A, a whole output vector




Axler, S. ***Linear Algebra Done Right***, 4th ed.: standard basis 2.27(a), p. 39; columns 3.50, p. 76. Springer, 2024.

# A matrix is a map

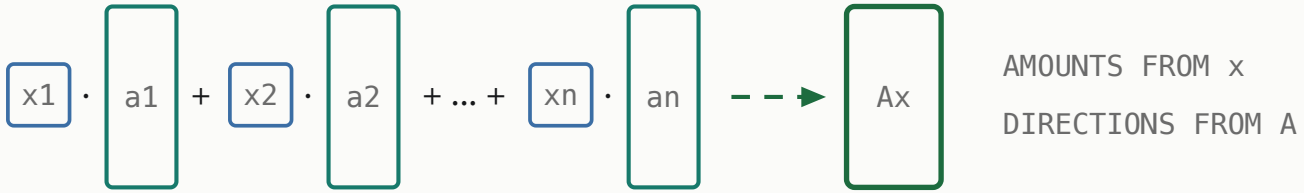
LINEAR COMBINATION OF COLUMNS

$$Ax = \sum_{j=1}^{n} x_j \, (Ae_j) = x_1 a_1 + \cdots + x_n a_n$$

$x_j$ coordinate j of x, used as an amount

$Ae_j$ column j, from the page before

$a_j$ column j of A, a whole direction



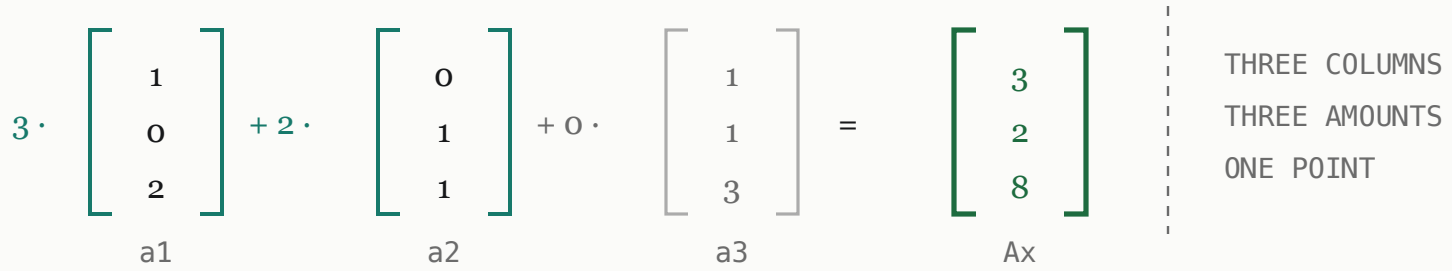

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.50 linear combination of columns, p. 76. Springer, 2024.

# A matrix is a map

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = 3a_1 + 2a_2 = \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix}$$

$a_1, a_2$ (1,0,2) and (0,1,1), the first two columns

3, 2, 0 the amounts asked of the three columns



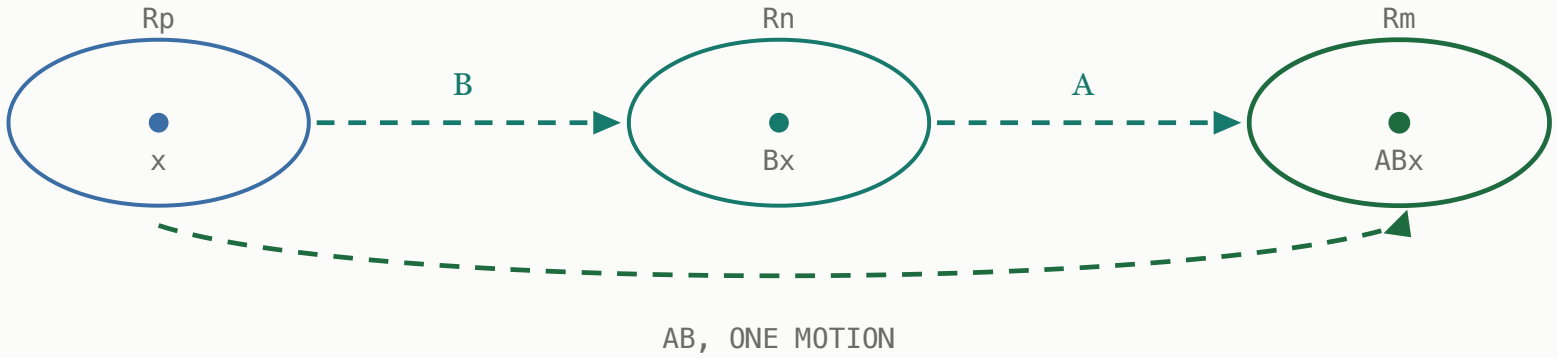
# A matrix is a map

DO B FIRST, THEN A $(AB)x = A(Bx)$

shapes $B \in \mathbb{R}^{n\times p}$, $A \in \mathbb{R}^{m\times n}$

Bx where B sends x first

A(Bx) where A then sends that




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.43 matrix of product of linear maps, p. 74. Springer, 2024.

# A matrix is a map

**TRANSPOSE** $(A^{\mathsf{T}})_{k,j} = A_{j,k}$

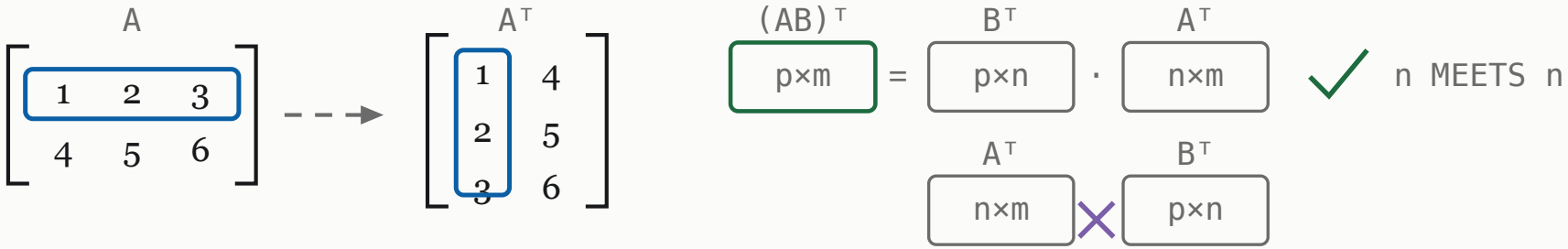
`row j` of A becomes column j of $A^{\mathsf{T}}$

`m × n` becomes n × m

**IT REVERSES A PRODUCT** $\boxed{(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}}$

$(AB)^{\mathsf{T}}$ p × m, the shape the answer must have

$B^{\mathsf{T}}A^{\mathsf{T}}$ p × n times n × m, so n meets n




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.54 transpose, p. 77; $(AC)^t = C^tA^t$, p. 78. Springer, 2024.

INTEGRATED WORKED EXAMPLE

# A matrix is a map

$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. Apply $B$, then $A$, to $x = (1, -1)^{\mathsf{T}}$; check the transpose.

$$AB = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

reason $\mathbb{R}^2 \to \mathbb{R}^3 \to \mathbb{R}^2$: the inner dimensions match.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# A matrix is a map

Continue; check the original map.

$$AB = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

reason $\mathbb{R}^2 \to \mathbb{R}^3 \to \mathbb{R}^2$: the inner dimensions match.

$$Bx = (1, -1, 0)^\mathsf{T}, \quad A(Bx) = (-1, -1)^\mathsf{T} = (AB)x$$

reason Following the intermediate vector checks the product.

$$(AB)^\mathsf{T} = B^\mathsf{T} A^\mathsf{T} = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

reason BA is 3 by 3; it describes a different trip.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

INDEPENDENT TRANSFER

# A matrix is a map

Keep $B$. Now $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$ and $x = (2, 1)^{\mathsf{T}}$. Find $AB$, its output and $(AB)^{\mathsf{T}}$.

$$AB = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$$

reason Column $j$ is $A(Be_j)$.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# A matrix is a map

Continue; check the original map.

$$AB = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$$

reason Column $j$ is $A(Be_j)$.

$$Bx = (2, 1, 3)^{\mathsf{T}}, \quad (AB)x = (8, -2)^{\mathsf{T}}$$

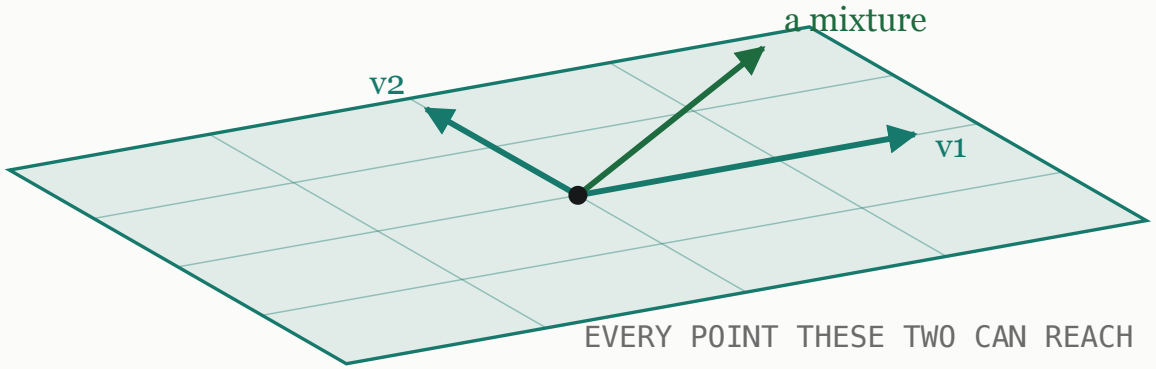
reason Check each output using the intermediate vector.

$$(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}} = \begin{bmatrix} 3 & -1 \\ 2 & 0 \end{bmatrix}$$

reason Transposition reverses the factor order.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

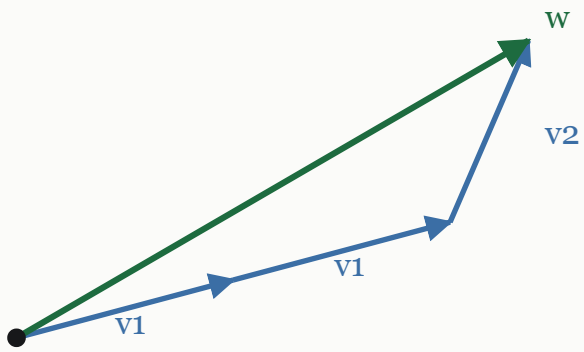
# Span and column space



# Span and column space

LINEAR COMBINATION $w = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$

- $v_i$ the directions you are handed
- $c_i$ any reals, negative and zero included
- w where those weights land you




TWO STEPS OF v1

ONE STEP OF v2

w = 2 v1 + v2

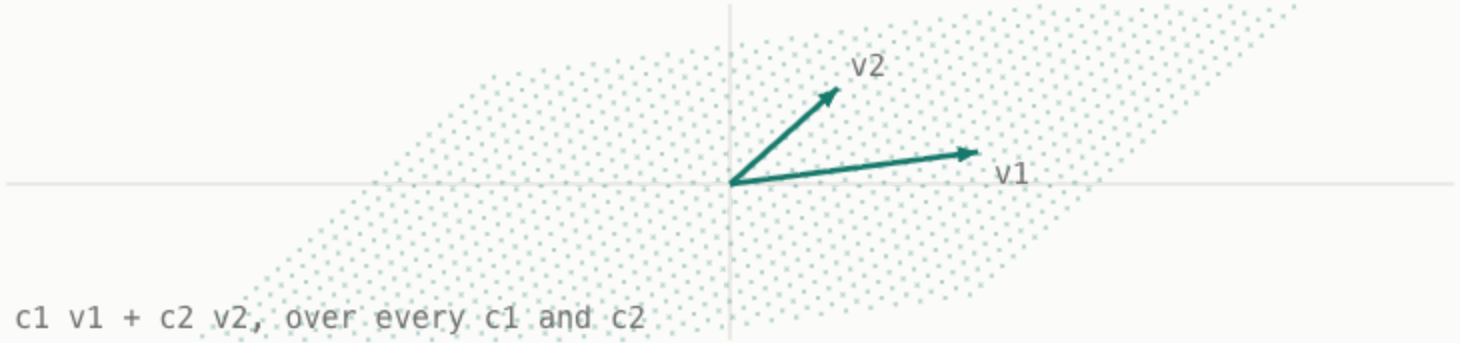
Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.2 linear combination, p. 28. Springer, 2024.

# Span and column space

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

**SPAN** $\operatorname{span}(v_1, \ldots, v_k) = \{c_1 v_1 + \cdots + c_k v_k \ : \ c_i \in \mathbb{R}\}$

{ } every w the weights can reach, all at once



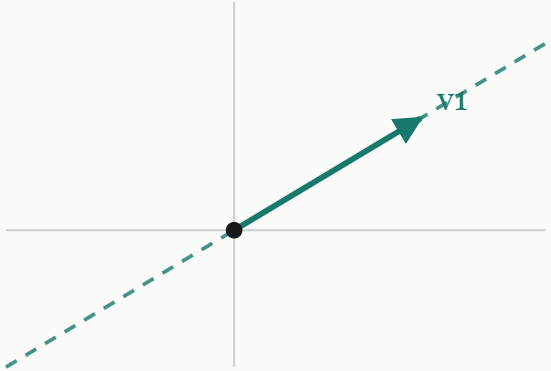

Let both weights roam: the plane fills in.

**WORKED EXAMPLE**

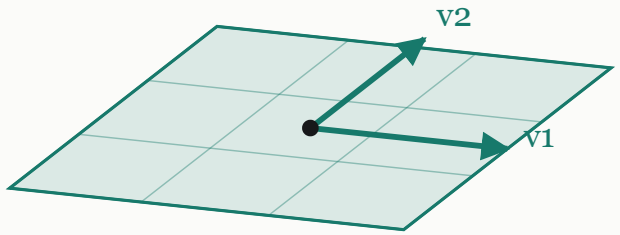
$v_1 = (1, 0, 2)$, $v_2 = (0, 1, 1)$: $3v_1 + 2v_2 = (3, 2, 8)$ is in the span.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.4 span, p. 29. Springer, 2024.

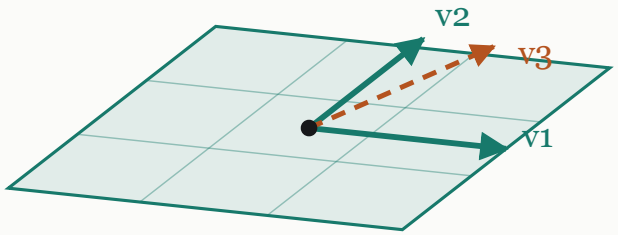
# Span and column space




ONE NONZERO VECTOR · A LINE




TWO INDEPENDENT ONES · A PLANE




A THIRD IN IT · SAME PLANE

$$\operatorname{span}(v_1) = \{c\, v_1\}$$
$$\operatorname{span}(v_1, v_2) = \{c_1 v_1 + c_2 v_2\}$$
$$\operatorname{span}(v_1, v_2, v_3) = \operatorname{span}(v_1, v_2)$$

`the conditions` dependent pair: a line, or $\{0\}$ if both zero

`v₃` already a mixture of the first two

Sanderson, G. ***Essence of Linear Algebra***, Chapter 2. 3Blue1Brown, 2016.

# Span and column space

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

$$b \in \text{span}(v_1, \ldots, v_k) \iff Vc = b \text{ has a solution}$$

- `V` the arrows as columns
- `c` the unknown weights
- `b` the target vector

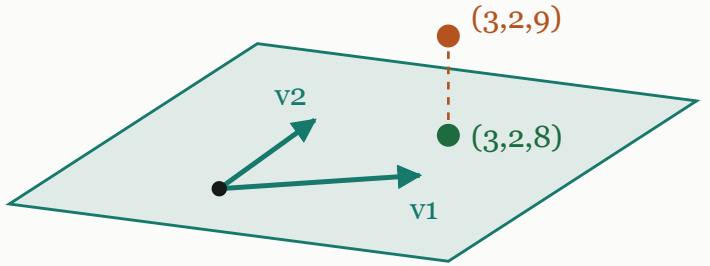
**IS (3,2,9) IN THE SPAN?**

$c_1(1,0,2) + c_2(0,1,1) = (3,2,9)$

row 1 gives $c_1 = 3$, row 2 gives $c_2 = 2$

row 3 is then forced: $2(3) + 2 = 8$, not 9

span of v1, v2




One lands, one misses.

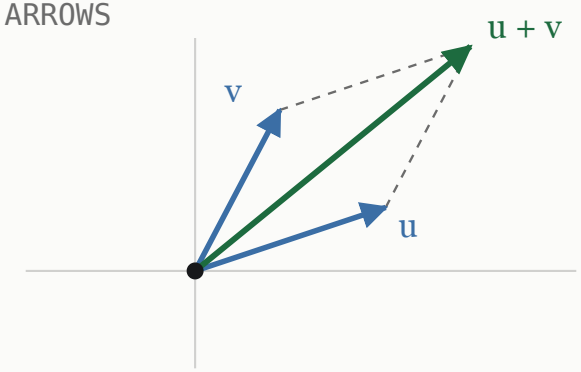
# Span and column space

A vector space is a set equipped with vector addition and scalar multiplication satisfying the vector-space axioms.

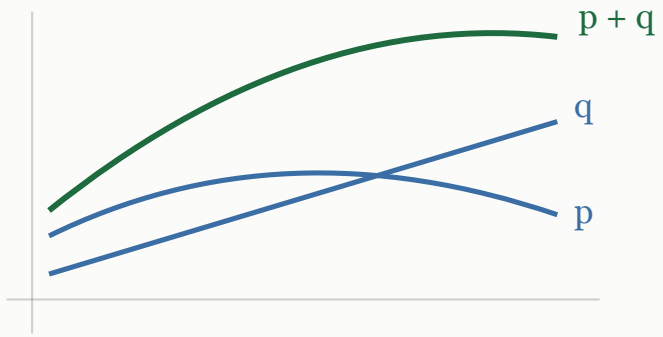
WHAT WE USE $$u, v \in V,\ \alpha, \beta \in \mathbb{R} \implies \alpha u + \beta v \in V$$

`closure` the one property this course leans on all term

ARROWS




POLYNOMIALS




MATRICES

$$\begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} + \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix}$$

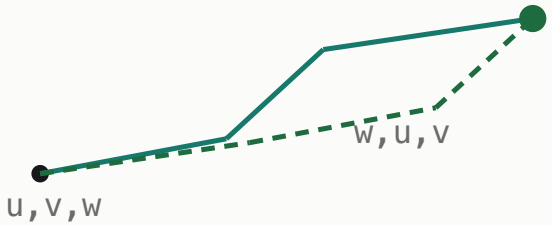
Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.20 vector space, p. 12. Springer, 2024.

# Span and column space

ADDITION

$$u+v=v+u \qquad (u+v)+w=u+(v+w)$$
$$v+0=v \qquad \exists w:\ v+w=0$$

`0 and` $w$ an origin, and an undo for every vector




ANY ORDER, SAME POINT

SCALING

$$1v=v \qquad \alpha(\beta v)=(\alpha\beta)v$$
$$\alpha(u+v)=\alpha u+\alpha v \qquad (\alpha+\beta)v=\alpha v+\beta v$$

`the payoff` obey all eight and every theorem here applies

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.20 vector space, p. 12. Springer, 2024.

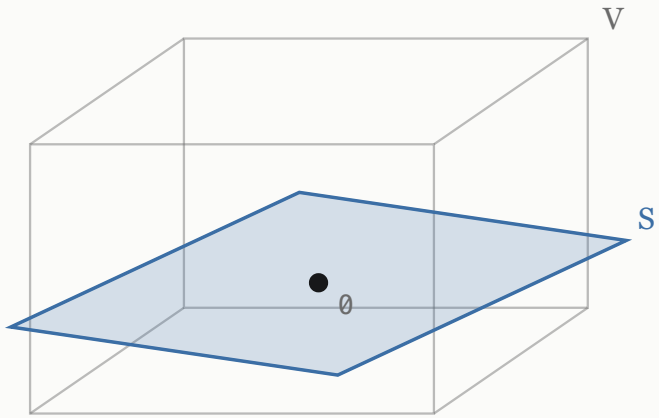
# Span and column space

A subset $S \subseteq V$ is a subspace if it is itself a vector space under the same operations.

EQUIVALENT TEST $\boxed{S \neq \varnothing \text{ and } u, v \in S,\ \alpha, \beta \in \mathbb{R} \Rightarrow \alpha u + \beta v \in S}$

`nonempty` plus closure already gives $0 \in S$: choose $\alpha = \beta = 0$

`next page` the same test, split into three checks

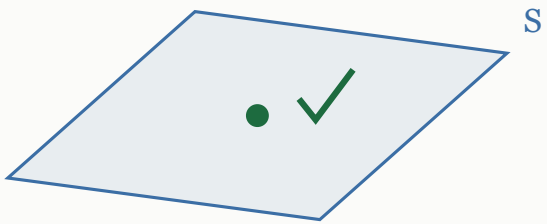



Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.33 subspace and 1.34 conditions for a subspace, p. 18. Springer, 2024.

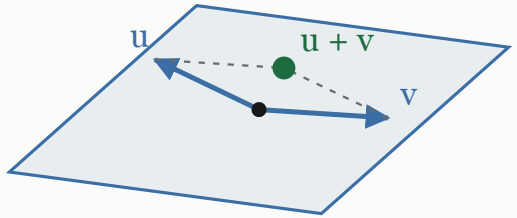
# Span and column space

CONDITIONS FOR A SUBSPACE $0 \in S; \qquad u + v \in S; \qquad \alpha u \in S$

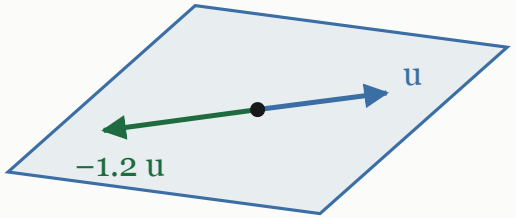
| | |
|---|---|
| `0` | the zero vector is a member |
| `u+v` | any two members add to a member |
| `alpha u` | any multiple stays in, negatives included |




0 IS IN S




u + v IS IN S



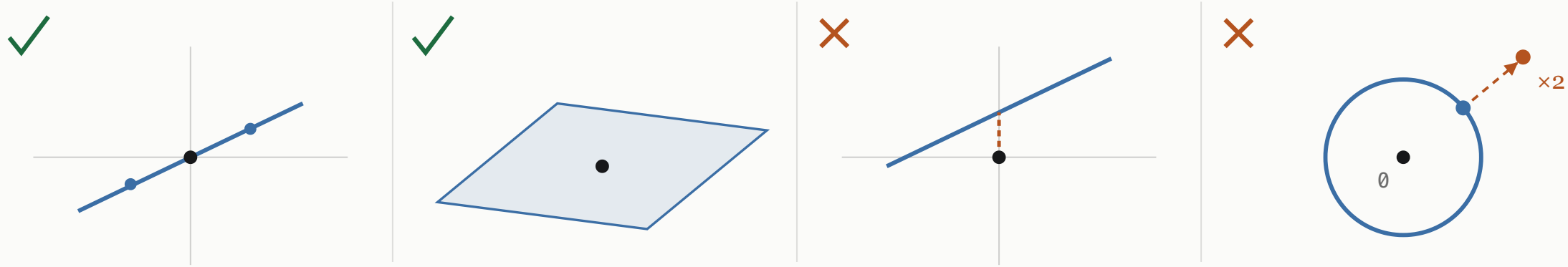

ANY MULTIPLE IS IN S

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.34 conditions for a subspace, p. 18. Springer, 2024.

# Span and column space

FASTEST DISQUALIFIER $0 \notin S \implies S$ is not a subspace

`0 ∈ S` necessary, never sufficient: closure still has to hold




LINE THROUGH 0 · PLANE THROUGH 0 · SHIFTED LINE · AFFINE · CIRCLE

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.35 example: subspaces, p. 19. Springer, 2024.

# Span and column space

CLAIM $v_1, \ldots, v_k \in V \implies \operatorname{span}(v_1, \ldots, v_k)$ is a subspace of $V$

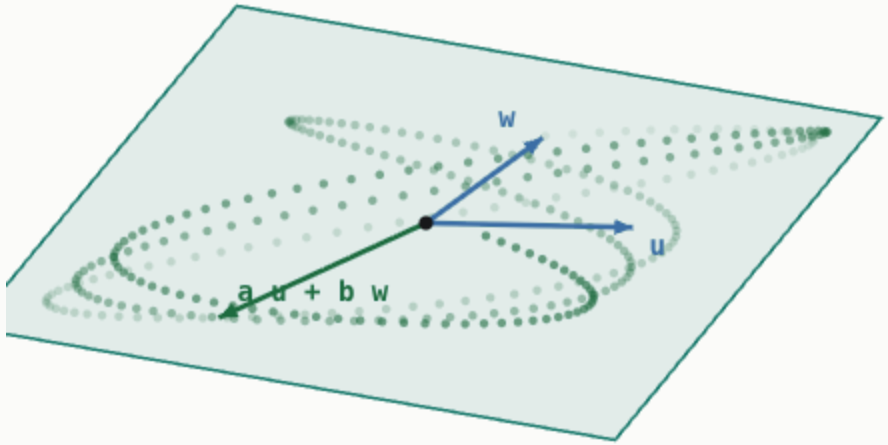
`S` every mixture of the list; the smallest subspace containing it

$$0 = \sum_{i=1}^{k} 0\, v_i \in S$$

`every c_i = 0` the origin is a member

$$\alpha u + \beta w = \sum_{i=1}^{k} (\alpha c_i + \beta d_i)\, v_i \in S$$

`u, w` any two members: $u = \sum_i c_i v_i$, $w = \sum_i d_i v_i$

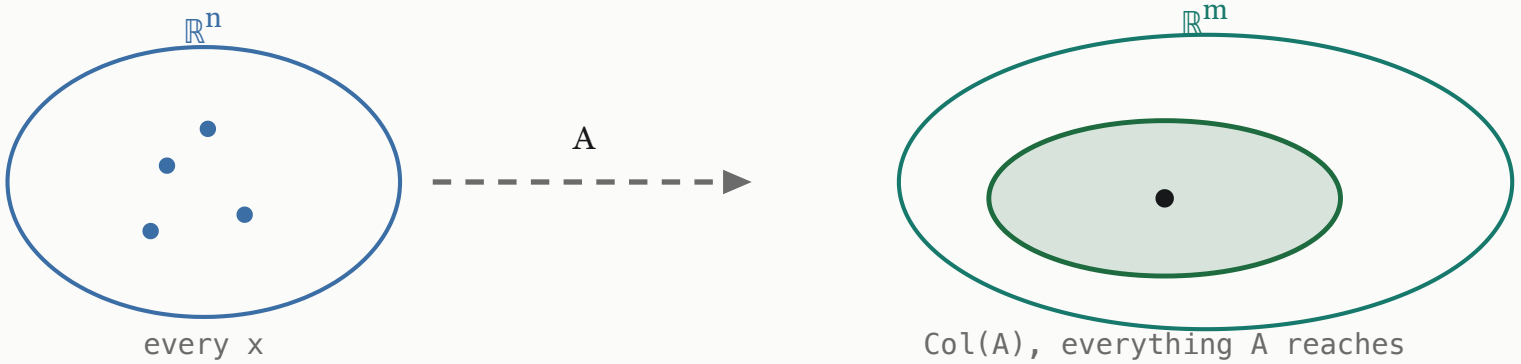



Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.6 span is the smallest containing subspace, p. 29. Springer, 2024.

# Span and column space

COLUMN SPACE $$\boxed{\mathrm{Col}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m}$$

| | |
|---|---|
| A | any $m \times n$ matrix; x runs over the whole of $\mathbb{R}^n$ |
| $\subseteq \mathbb{R}^m$ | a subspace of the codomain; if $m = n$ the roles still differ |
| also called | the image of A, or the range of A |




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range and 3.18 the range is a subspace, p. 61. Springer, 2024.

# Span and column space

DEFINITION

$$\mathrm{Col}(A) = \{Ax : x \in \mathbb{R}^n\}$$

`x` runs over every input

COLUMN VIEW

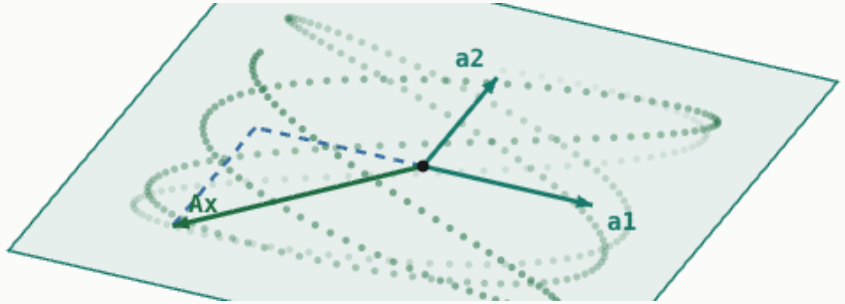
$$Ax = \sum_{j=1}^{n} x_j \, a_j$$

$a_j$ column j of A

⇓

$$\boxed{\mathrm{Col}(A) = \mathrm{span}(a_1, \ldots, a_n)}$$

`one set` reachable outputs and mixtures of the columns are the same thing




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61; 3.50 linear combination of columns, p. 76. Springer, 2024.

# Span and column space

QUESTION $Ax = b$ ?

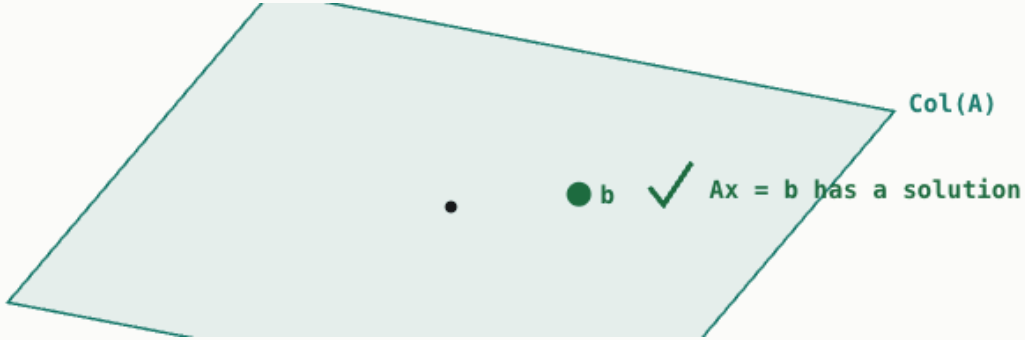
`b` a target on the output side

$$Ax = b \text{ has a solution} \iff \exists\, x :\ Ax = b \iff b \in \{Ax : x \in \mathbb{R}^n\}$$

`∃ x` the whole question is whether one input reaches b

$$\boxed{Ax = b \text{ is solvable} \iff b \in \mathrm{Col}(A)}$$

`nothing new` the right side is the definition of Col(A), read backwards




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61. Springer, 2024.

# Span and column space

SOLVE $$c_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ y_3 \end{bmatrix}$$

row 1 $c_1 + 0 = 3$, so $c_1 = 3$

row 2 $0 + c_2 = 2$, so $c_2 = 2$

row 3 $2c_1 + c_2 = 2(3) + 2 = 8$ must equal $y_3$

TARGET 1

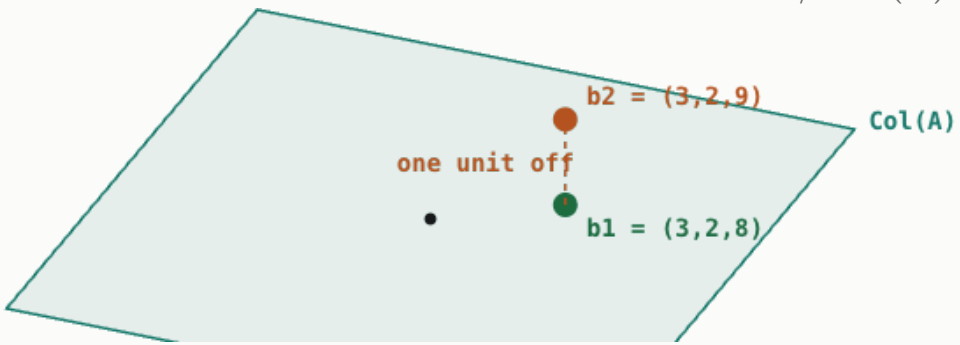
$y_3 = 8 : \; 8 = 8$ ✓

so $b_1 = 3a_1 + 2a_2 \in \mathrm{Col}(A)$

TARGET 2

$y_3 = 9 : \; 9 \neq 8$ ×

so no $c$ works: $b_2 \notin \mathrm{Col}(A)$




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61. Springer, 2024.

# Span and column space

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\boxed{Ax = \sum_j x_j \, a_j}$$

$x_j$ the amount taken of column j

$$\Downarrow$$

$$\boxed{\mathrm{Col}(A) = \mathrm{span}(a_1, \ldots, a_n)}$$

span every mixture, so every reachable output

$$\Downarrow$$

$$\boxed{Ax = b \text{ solvable} \iff b \in \mathrm{Col}(A)}$$

our A $a_3 = a_1 + a_2$, so the third column adds nothing to the span

> When do different columns carry independent information?

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.15 linearly independent, p. 32. Springer, 2024.

INTEGRATED WORKED EXAMPLE

# Span and column space

$S = \{y \in \mathbb{R}^3 : y_3 = y_1 + 2y_2\}$. Prove $S$ is a subspace, give spanning vectors, and test $b = (2, -1, 0)^\mathsf{T}$.

$$y = (s, t, s + 2t)^\mathsf{T} = s(1, 0, 1)^\mathsf{T} + t(0, 1, 2)^\mathsf{T}$$

reason Every member is a combination, and every combination satisfies the equation.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# Span and column space

Continue; check the original map.

$$y = (s, t, s + 2t)^{\mathsf{T}} = s(1, 0, 1)^{\mathsf{T}} + t(0, 1, 2)^{\mathsf{T}}$$

reason Every member is a combination, and every combination satisfies the equation.

$$S = \operatorname{span}\{(1, 0, 1)^{\mathsf{T}}, (0, 1, 2)^{\mathsf{T}}\}$$

reason A span contains zero and is closed under linear combinations.

$$b = 2(1, 0, 1)^{\mathsf{T}} - (0, 1, 2)^{\mathsf{T}} \in S$$

reason The third coordinate checks: 2 minus 2 is zero.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

INDEPENDENT TRANSFER

# Span and column space

$T = \{y : y_3 = 2y_1 - y_2\}$. Find spanning vectors. Is $(1, 3, 0)^\mathsf{T}$ in $T$? Is $y_3 = 2y_1 - y_2 + 1$ a subspace?

$$T = \operatorname{span}\{(1, 0, 2)^\mathsf{T}, (0, 1, -1)^\mathsf{T}\}$$

reason Use the first two coordinates as free coefficients.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# Span and column space

Continue; check the original map.

$$T = \operatorname{span}\{(1, 0, 2)^{\mathsf{T}}, (0, 1, -1)^{\mathsf{T}}\}$$

reason Use the first two coordinates as free coefficients.

$$2(1) - 3 = -1 \neq 0$$

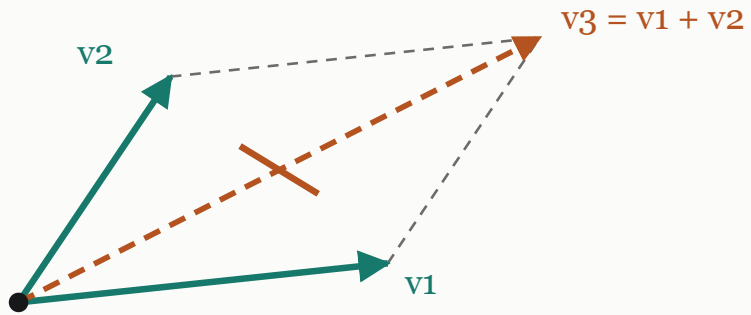
reason The target fails the defining equation.

$$0 \neq 2(0) - 0 + 1$$

reason The shifted set misses zero and is not a subspace.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

# Null Space and Linear Independence




ENOUGH TO REACH?

OR ONE TOO MANY?

# Null Space and Linear Independence

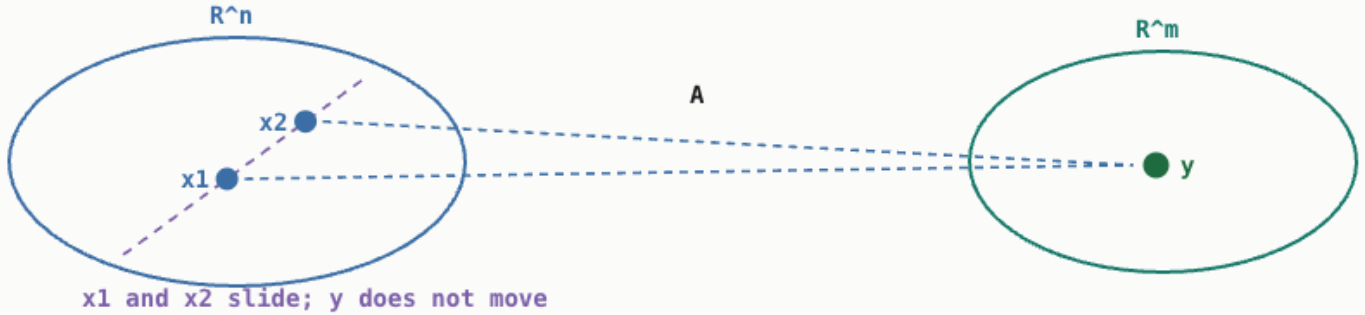
SO FAR $\mathrm{Col}(A) = \text{all reachable outputs}$

settled which targets can be hit, and by which mixtures

NOW ASK $\boxed{Ax_1 = Ax_2, \qquad x_1 \neq x_2}$

one output but more than one input that reaches it

> How can two different inputs produce the same output?




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.14 injective, p. 60; 3.11 null space, p. 59. Springer, 2024.

# Null Space and Linear Independence

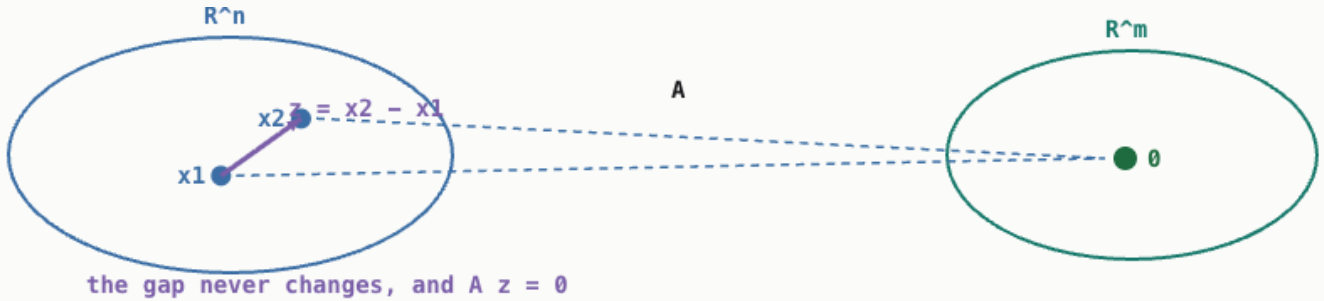
$$Ax_1 = Ax_2$$
$$Ax_1 - Ax_2 = 0$$
$$A(x_1 - x_2) = 0$$

`last step` linearity, and nothing else, is what pulls A outside

THE DIFFERENCE $\boxed{z = x_1 - x_2 \implies Az = 0}$

`z` nonzero, since $x_1 \neq x_2$, and A sends it to zero

> Every collision $Ax_1 = Ax_2$ is a solution $z = x_1 - x_2$ of $Ax = 0$.




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.1 linear map, p. 52; 3.11 null space, p. 59. Springer, 2024.

# Null Space and Linear Independence

HOMOGENEOUS SYSTEM $\boxed{Ax = 0}$

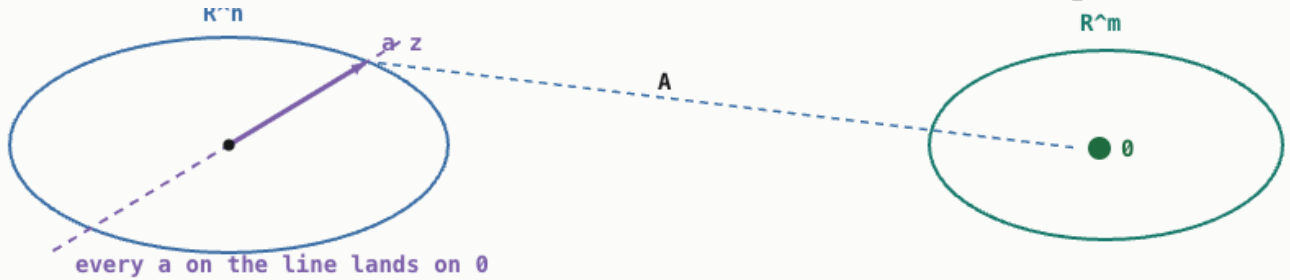
`homogeneous` the right-hand side is the zero vector

`x = 0` always a solution, and the uninteresting one

> Are there nonzero solutions?

ONE GIVES A LINE $\boxed{Az = 0,\ z \neq 0 \implies A(\alpha z) = 0\ \forall \alpha \in \mathbb{R}}$

`αz` one invisible direction, so a whole line of invisible inputs




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.26 homogeneous system of linear equations, p. 65. Springer, 2024.

# Null Space and Linear Independence

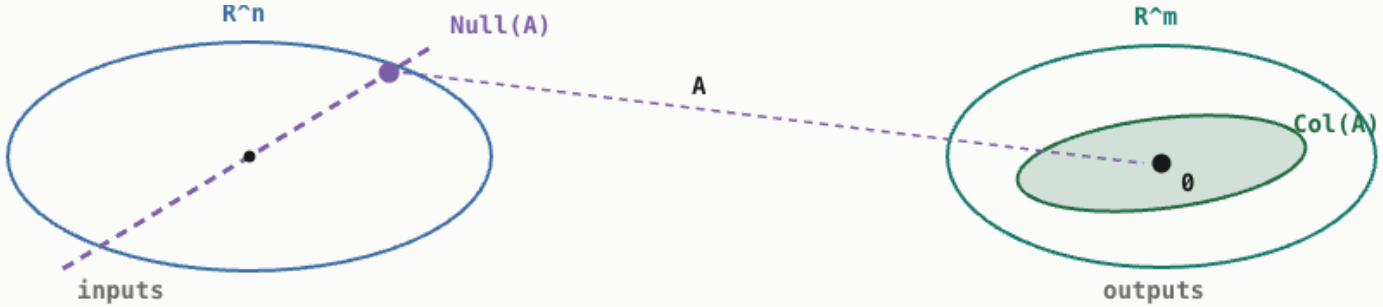
NULL SPACE $\boxed{\text{Null}(A) = \{x \in \mathbb{R}^n : Ax = 0\}}$

also called the kernel of A, written $\ker(A)$

$Ax = 0$ the inputs A cannot see

SHAPE $\boxed{\text{Null}(A) \subseteq \mathbb{R}^n}$ against $\text{Col}(A) \subseteq \mathbb{R}^m$

input side Null(A) lives where x lives, Col(A) where Ax lives




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space and 3.13 the null space is a subspace, p. 59. Springer, 2024.

# Null Space and Linear Independence

$$u, v \in \text{Null}(A) \implies Au = 0,\ Av = 0$$

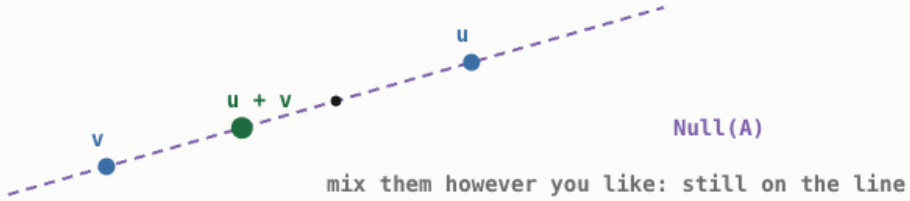
and $A0 = 0$, so the origin is in $\text{Null}(A)$ to begin with

$$A(\alpha u + \beta v) = \alpha Au + \beta Av = 0$$

so $\alpha u + \beta v \in \text{Null}(A)$ for every $\alpha, \beta$

SUBSPACE $\boxed{\text{Null}(A) \text{ is a subspace of } \mathbb{R}^n}$

flat in $\mathbb{R}^3$: $\{0\}$, a line, a plane, or all of $\mathbb{R}^3$




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.13 the null space is a subspace, p. 59. Springer, 2024.

# Null Space and Linear Independence

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

OUR MATRIX $a_3 = a_1 + a_2 \implies a_1 + a_2 - a_3 = 0$

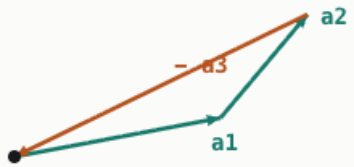
read it a mixture of the columns, weights $1, 1, -1$, giving zero

$$A \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = 0 \implies \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \in \mathrm{Null}(A)$$

and next: solve for the whole set

THE WHOLE SET $\boxed{\mathrm{Null}(A) = \mathrm{span}\left\{(1, 1, -1)^{\mathsf{T}}\right\}}$

a line through the origin




a1 + a2 − a3 walks back to 0

so (1, 1, −1) is in Null(A)

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Null Space and Linear Independence

$$\text{Ax = 0} \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

row 1 $1x_1 + 0x_2 + 1x_3 = 0$, so $x_1 + x_3 = 0$

row 2 $0x_1 + 1x_2 + 1x_3 = 0$

row 3 $2x_1 + x_2 + 3x_3 = 0$ is twice row 1 plus row 2

SOLVE $x_3 = t:\ x_1 = -t,\ x_2 = -t,\ x = t(-1, -1, 1)^{\mathsf{T}}$

why one free unknown

NULL SPACE $\boxed{\text{Null}(A) = \text{span}\{(-1, -1, 1)^{\mathsf{T}}\}}$

check $-1 + 1,\ -1 + 1,\ -2 - 1 + 3$: all zero

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Null Space and Linear Independence

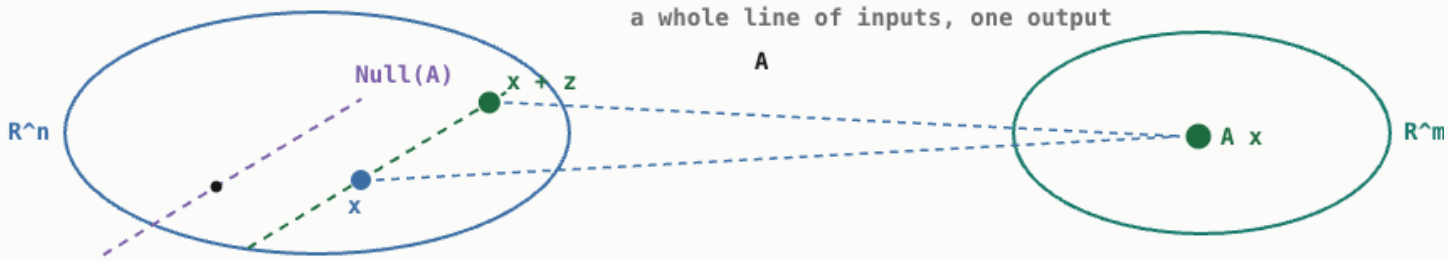
$$z \in \text{Null}(A) \implies Az = 0 \implies A(x+z) = Ax + Az = Ax$$

`x` any input at all, not a special one

WHAT IT MEANS $\boxed{A(x+z) = Ax, \qquad z \in \text{Null}(A)}$

`x + z` for our matrix, a line of inputs sharing one output

> Null-space directions are invisible to the map.




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.1 linear map, p. 52; 3.11 null space, p. 59. Springer, 2024.

# Null Space and Linear Independence

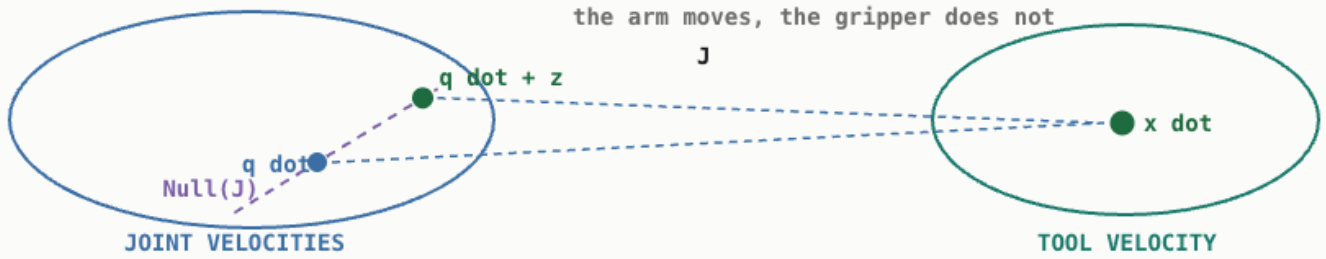
ON A ROBOT $\dot{x} = J(q)\,\dot{q}$

$\dot{q}$ joint velocities, what you command

$\dot{x}$ end-effector velocity, what the world sees

REDUNDANCY $\boxed{z \in \text{Null}(J(q)) \implies J(q)(\dot{q} + z) = J(q)\dot{q}}$

z a joint-velocity direction that produces no end-effector velocity




Lynch, K. M. and Park, F. C. ***Modern Robotics***, §6.3 inverse velocity kinematics, p. 233. Cambridge University Press, 2017 preprint.

# Null Space and Linear Independence

LINEAR INDEPENDENCE $$c_1 v_1 + \cdots + c_k v_k = 0 \implies c_1 = \cdots = c_k = 0$$

`read it` the all-zero weights are the only ones that cancel

SAME THING $$\sum_{i=1}^{k} c_i v_i = 0 \implies c = 0$$

`c` the whole list of weights, read as one vector




DEPENDENT · THE WALK CLOSES

INDEPENDENT · IT NEVER CLOSES

Left, a nonzero walk comes home. Right, none does.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.15 linearly independent, p. 32. Springer, 2024.

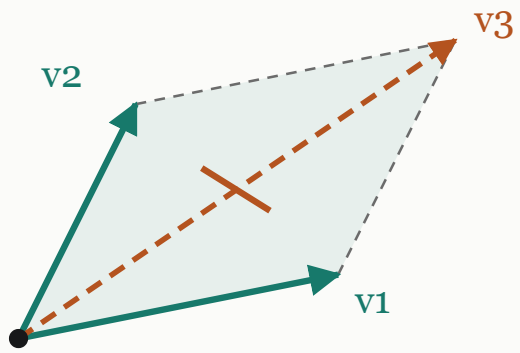
# Null Space and Linear Independence

LINEAR DEPENDENCE $$\boxed{\exists\, c \neq 0 :\ c_1 v_1 + \cdots + c_k v_k = 0}$$

$c \neq 0$ at least one weight is nonzero; say it is $c_j$

SOLVE FOR THAT ONE $$\boxed{v_j = -\frac{1}{c_j} \sum_{i \neq j} c_i v_i}$$

so $v_j$ is built from the others, and deleting it leaves the span unchanged




DELETE v3

SAME PLANE

Strike the diagonal and the plane is untouched.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.17 linearly dependent and 2.19 linear dependence lemma, p. 33. Springer, 2024.

# Null Space and Linear Independence

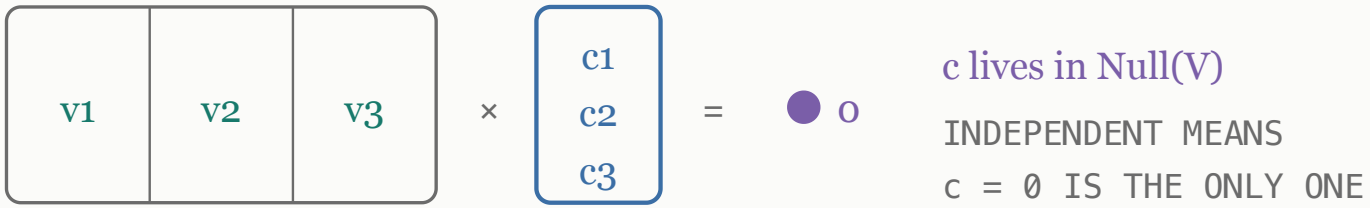
STACK THEM $V = [\, v_1 \ \cdots \ v_k \,], \qquad Vc = \sum_i c_i v_i$

`Vc` the matrix times its weights

THE ANCHOR

$$\begin{aligned} v_1, \ldots, v_k \text{ independent} &\iff \text{Null}(V) = \{0\} \\ v_1, \ldots, v_k \text{ dependent} &\iff \text{Null}(V) \neq \{0\} \end{aligned}$$

`one question` redundancy in a list is a null space




c lives in Null(V)

INDEPENDENT MEANS
c = 0 IS THE ONLY ONE

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.15 linearly independent, p. 32; 3.15 injectivity, p. 60. Springer, 2024.

# Null Space and Linear Independence

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

OUR MATRIX $a_3 = a_1 + a_2 \implies A(1, 1, -1)^{\mathsf{T}} = 0$

one equation the weights $1, 1, -1$ cancel the three columns

NULL SPACE

$$(1, 1, -1)^{\mathsf{T}} \in \mathrm{Null}(A)$$

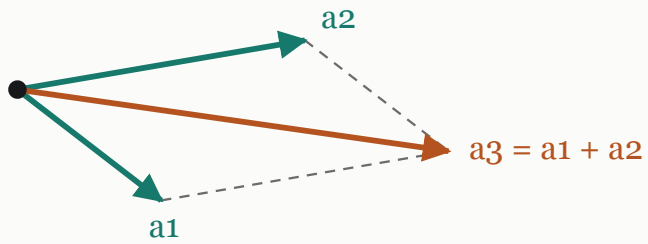
reading an input A cannot see

LINEAR DEPENDENCE

$$a_1, a_2, a_3 \text{ dependent}$$

reading a column you may delete

> Same equation, two interpretations.




THE THIRD COLUMN
BUYS NOTHING

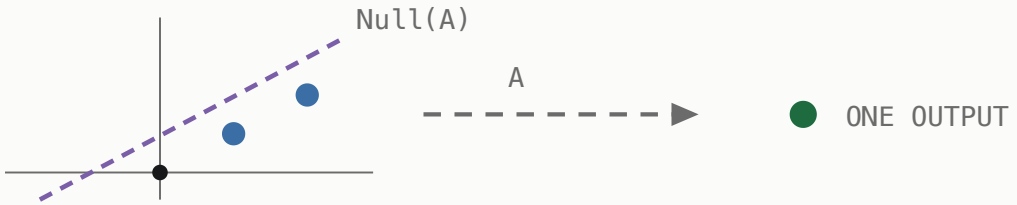
Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.17 linearly dependent, p. 33; 3.11 null space, p. 59. Springer, 2024.

# Null Space and Linear Independence

$$\begin{aligned} \text{Null}(A) &= \{x : Ax = 0\} \\ z \in \text{Null}(A) &\implies A(x+z) = Ax \\ \text{Null}(A) \neq \{0\} &\implies \text{information is lost} \end{aligned}$$

`the chain` definition, meaning, cost

FOR THE COLUMNS $\text{Null}(A) = \{0\} \iff \text{columns independent}$

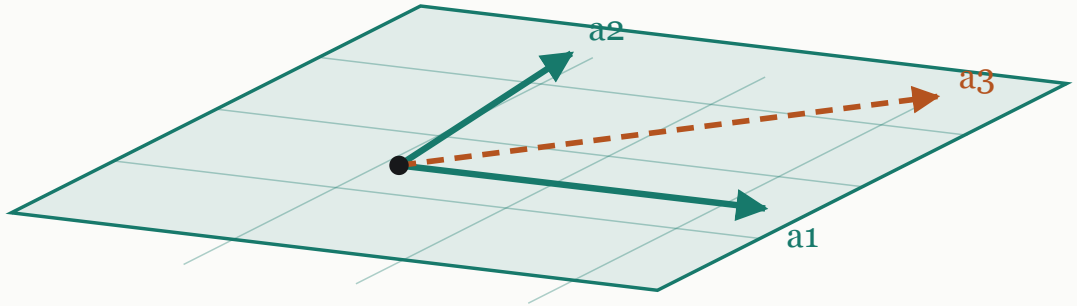



`settled` whether a direction is redundant

> How many independent directions?

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

# Basis, Dimension, and Rank




HOW MANY

SURVIVE?

# Basis, Dimension, and Rank

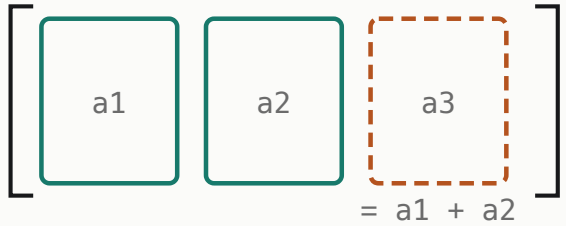
$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

WE KNOW $\mathrm{Col}(A)$ what A can produce, $\mathrm{Null}(A)$ what A cannot see

and how to tell whether a direction is redundant

NOW ASK **How many independent directions are needed?**

our A $a_3 = a_1 + a_2$: three columns, but how many directions?



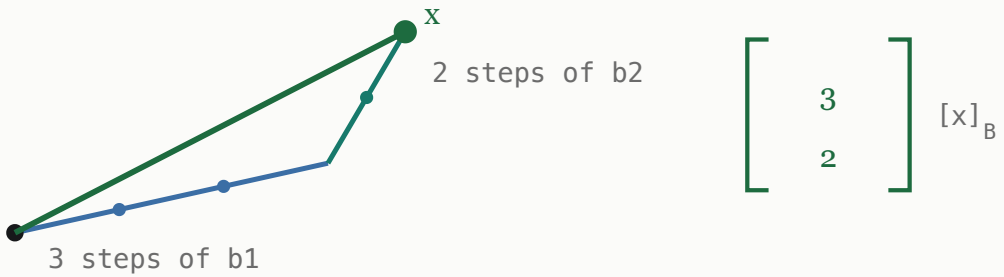

THREE COLUMNS

HOW MANY DIRECTIONS?

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

# Basis, Dimension, and Rank

**BASIS** $\boxed{\text{Basis} = \text{Spanning} + \text{Linear Independence}}$




`spans` $\operatorname{span}(b_1, \ldots, b_k) = V$

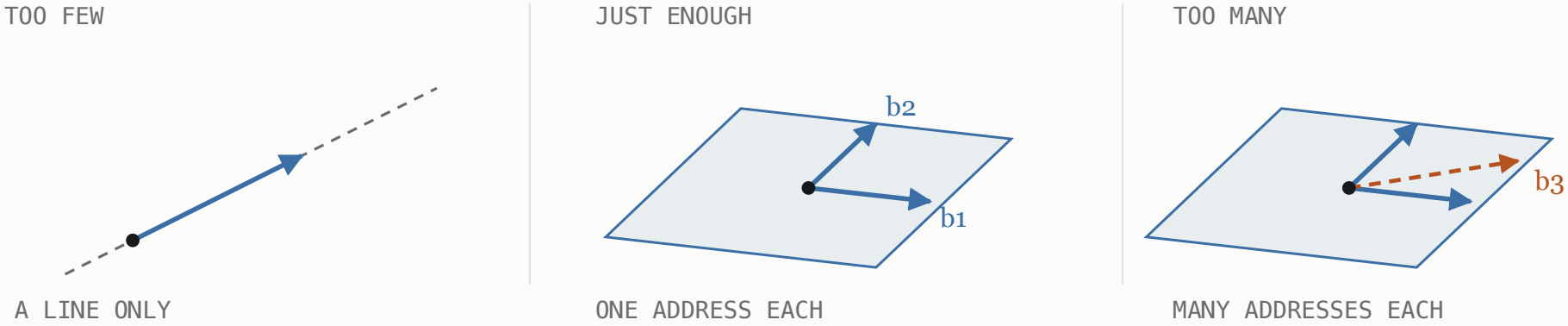
`independent` $c_1 b_1 + \cdots + c_k b_k = 0 \Rightarrow c_1 = \cdots = c_k = 0$

**SO** enough to build everything, with nothing spare

`both halves` either one alone is not a basis

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

# Basis, Dimension, and Rank




Too few cannot reach; too many give several addresses.

**BASIS** enough, but not too many

`too few` here independent, but does not reach everything

`too many` reaches everything, but carries a repeat

**WORKED EXAMPLE**

In $\mathbb{R}^2$: $\{(1,0)\}$ is too few, $\{(1,0),(1,1)\}$ is a basis, adding $(2,1)$ is too many.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

# Basis, Dimension, and Rank

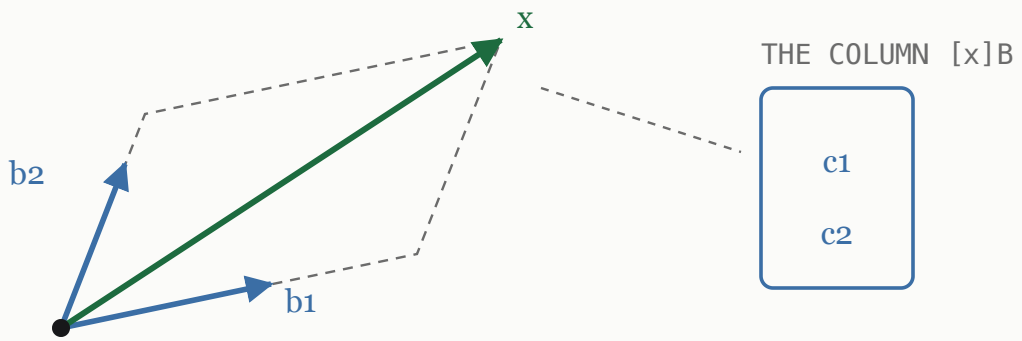
**UNIQUE WEIGHTS** $x = c_1 b_1 + \cdots + c_k b_k$ in exactly one way

**why** independence: two different weightings would differ by a nonzero mix that vanishes

**COORDINATES**

$$[x]_B = (c_1, \ldots, c_k)^{\mathsf{T}}, \qquad x = B\,[x]_B$$

**one address** the vector does not change; only its coordinates do




Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.28 criterion for basis, p. 39; 3.73 matrix of a vector, p. 88. Springer, 2024.

# Basis, Dimension, and Rank

**DIMENSION** $\dim V = \text{the number of vectors in any basis of } V$




one more independent direction each time

`any` every basis of the same space has the same length

**READING** dimension counts independent degrees of freedom

`finite` the vector spaces considered here have finite bases

$$\dim\{0\} = 0$$

`zero space` the empty list is a basis of it

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.35 dimension, p. 44. Springer, 2024.

# Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$




BASIS b BASIS d

One plane, two bases, both of size two.

SAME SIZE ALWAYS

different bases, different arrows, the same number of them

`this plane` every basis of it has exactly two vectors

`so` dimension belongs to the space, not to a choice

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.35 dimension, p. 44. Springer, 2024.

# Basis, Dimension, and Rank

**REDUCE** $$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow{r_3 \leftarrow r_3 - 2r_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{r_3 \leftarrow r_3 - r_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

`step 1` $(2,1,3) - 2(1,0,1) = (0,1,1)$

`step 2` $(0,1,1) - (0,1,1) = (0,0,0)$

`pivots` columns 1 and 2

**WHY R** $R = EA$, $E$ invertible : $Ax = 0 \iff Rx = 0$

`so` $\text{Null}(A) = \text{Null}(R)$

Strang, G. ***18.06***, Lecture 7: pivot variables and special solutions. MIT OCW, 2010.

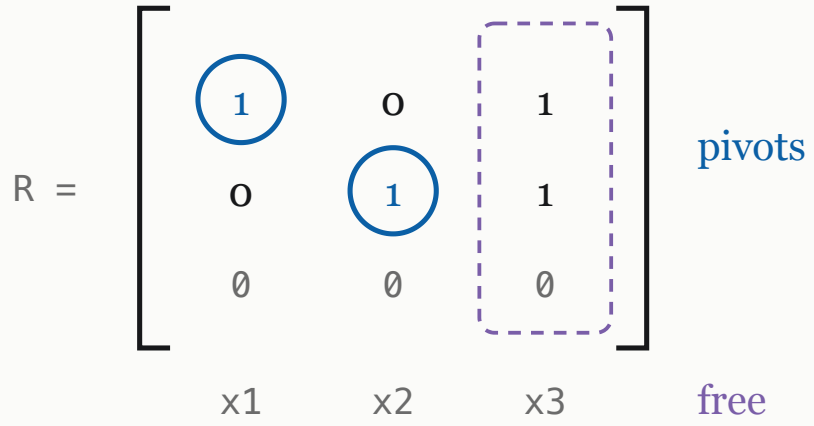
# Basis, Dimension, and Rank

**PIVOT** the leading nonzero entry of a nonzero row of $R$

`in R` pivot positions are read off the echelon form, not off $A$

`pivot variable` a variable whose column carries a pivot

`free variable` a variable whose column does not




**HERE** $x_1, x_2$ pivot; $x_3$ free

`count` 3 variables, 2 pivots, 1 free

Strang, G. ***18.06***, Lecture 7: pivot variables and special solutions. MIT OCW, 2010.

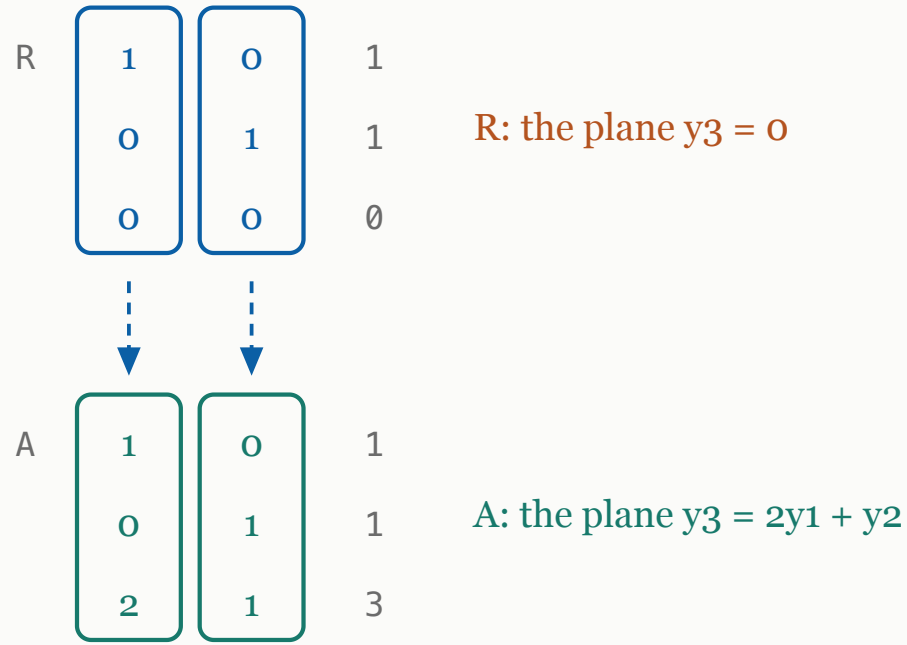
# Basis, Dimension, and Rank

**PIVOT COLUMNS** | the pivot columns of $A$ are a basis for $\mathrm{Col}(A)$

here $\mathrm{Null}(A) = \mathrm{Null}(R)$, while $\mathrm{Col}(A) \neq \mathrm{Col}(R)$

R's columns $(1,0,0)^{\mathsf{T}}, (0,1,0)^{\mathsf{T}}$ span $y_3 = 0$; $a_1 = (1,0,2)^{\mathsf{T}}$ has $y_3 = 2$

A's columns $a_1, a_2$ independent, and $a_3 = a_1 + a_2$, so they span $\mathrm{Col}(A)$




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.52 column rank, p. 77. Springer, 2024.

# Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

**ROWS OF R** $x_1 + x_3 = 0, \qquad x_2 + x_3 = 0, \qquad 0 = 0$

`legal` $\mathrm{Null}(A) = \mathrm{Null}(R)$

`free` column 3 has no pivot: $x_3 = t$

**SOLVE** $x_1 = -t, \quad x_2 = -t \implies x = (-t, -t, t)^{\mathsf{T}} = t(-1, -1, 1)^{\mathsf{T}}$

$t = 1$ on $A$: $-1 + 1,\ -1 + 1,\ -2 - 1 + 3$, all zero

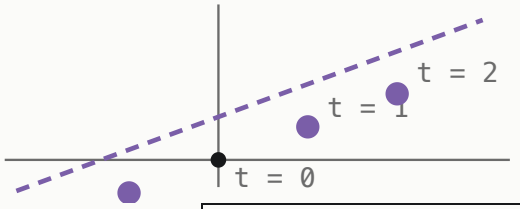
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Basis, Dimension, and Rank

ALL SOLUTIONS $\mathrm{Null}(A) = \{\, t(-1,-1,1)^{\mathsf{T}} : \; t \in \mathbb{R} \,\}$

why every solution is $t$ times one vector, nothing else

so all multiples of one vector: a span




BASIS AND DIMENSION $\mathrm{Null}(A) = \mathrm{span}\{(-1,-1,1)^{\mathsf{T}}\}, \qquad \dim \mathrm{Null}(A) = 1$

basis one nonzero vector is independent, hence a basis

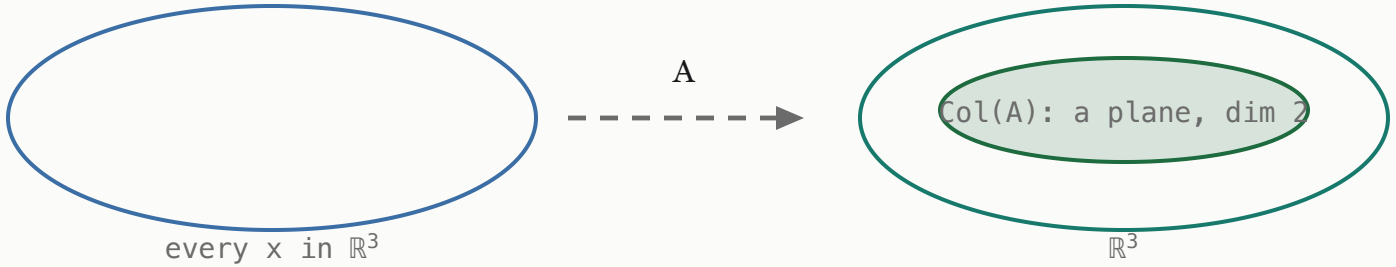
count one free variable, one basis vector: $\dim \mathrm{Null}(A)$ = free variables

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59; 2.26 basis, p. 39. Springer, 2024.

# Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

**RANK** $\boxed{\operatorname{rank}(A) = \dim \operatorname{Col}(A)}$




`our A` $\dim \operatorname{Col}(A) = 2$, so $\operatorname{rank}(A) = 2$

**TO COMPUTE** $\boxed{\operatorname{rank}(A) = \text{the number of pivots}}$

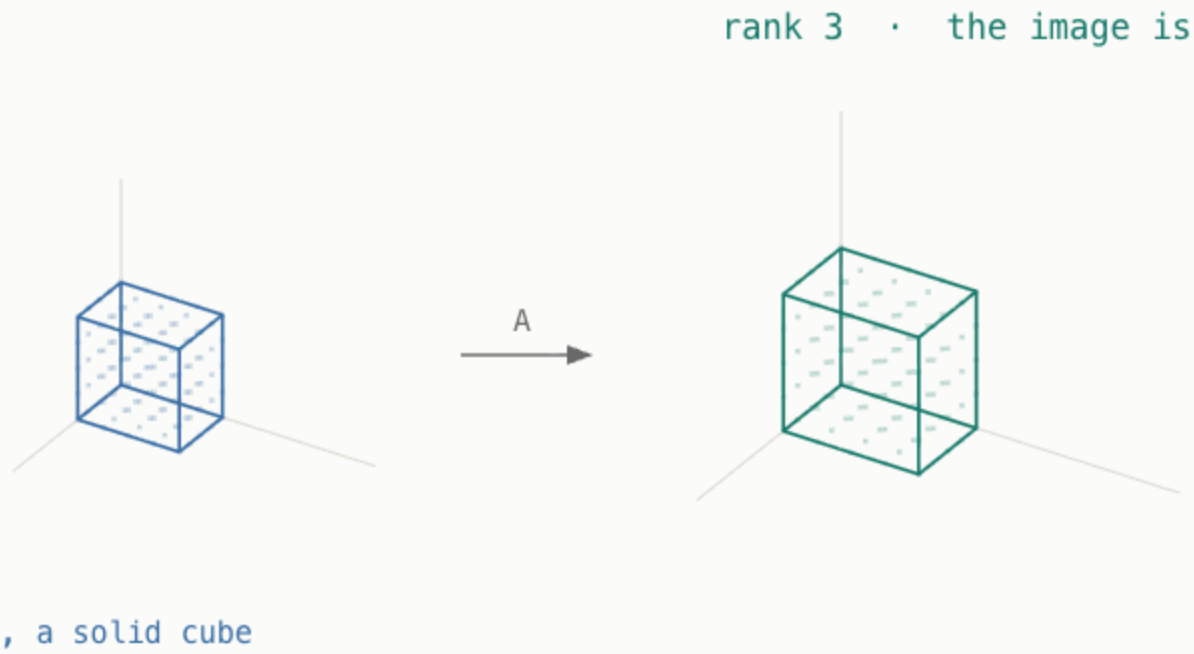
`reading` how many independent output directions A can produce

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.58 rank, p. 79; 3.52 column rank, p. 77. Springer, 2024.

# Basis, Dimension, and Rank

INTO THREE-SPACE $r = 3$ volume; $r = 2$ plane; $r = 1$ line; $r = 0$ the point $\{0\}$

`rank` the dimension of the reachable output space



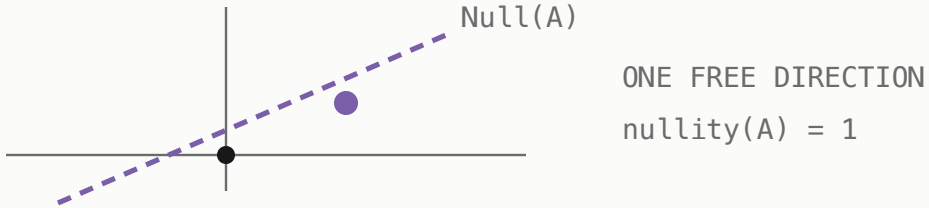

The input cube never changes. Each direction the matrix loses costs the image a whole dimension.

Sanderson, G. ***Essence of Linear Algebra***, ch. 7. 3Blue1Brown, 2016.

# Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

NULLITY $\boxed{\text{nullity}(A) = \dim \text{Null}(A)}$




our A $\text{Null}(A) = \text{span}\{(-1, -1, 1)^T\}$, so $\text{nullity}(A) = 1$

READING independent input directions lost by $A$

the pair rank counts what survives, nullity counts what disappears

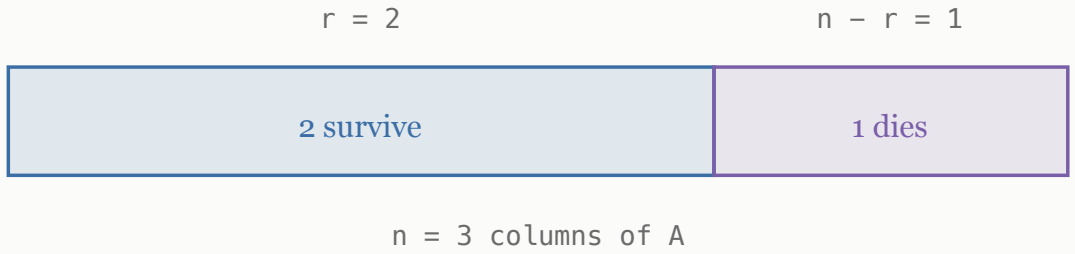
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

RANK–NULLITY $\boxed{\operatorname{rank}(A) + \operatorname{nullity}(A) = n}$

| | |
|---|---|
| `n` | the number of columns, the input dimension |
| `reach` | $\dim \operatorname{Col}(A) = r$: directions that survive |
| `loss` | $\dim \operatorname{Null}(A) = n - r$: directions that disappear |
| `our A` | $2 + 1 = 3$ |
| `a choice` | $n - r$ basis vectors in $\operatorname{Null}(A)$, $r$ whose images span $\operatorname{Col}(A)$ |



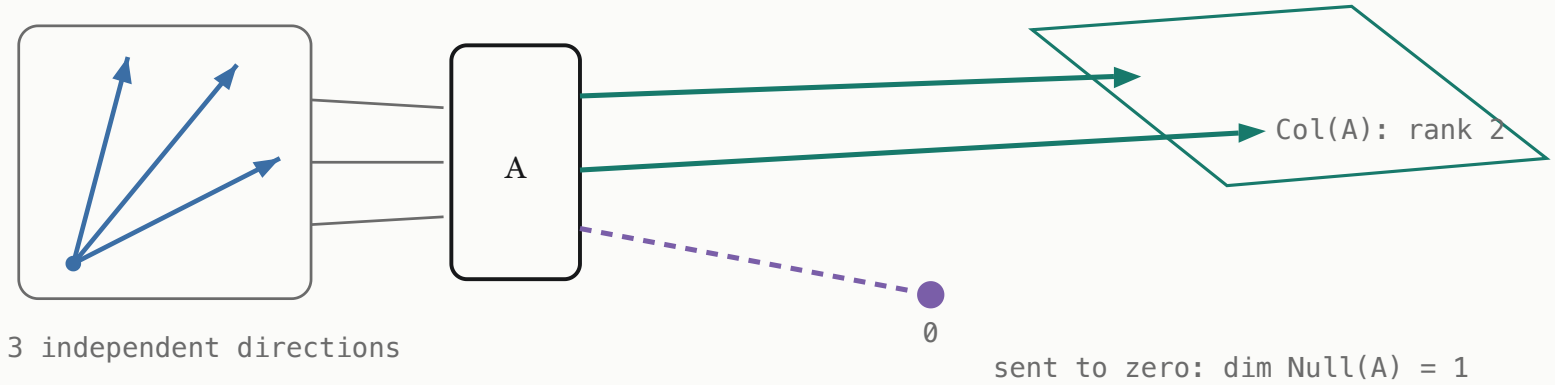

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

# Basis, Dimension, and Rank

THE MAP $A : \mathbb{R}^n \to \mathbb{R}^m, \qquad n = r + (n - r)$

`r` directions that survive

`n − r` directions that vanish




Two survivors, one casualty, three directions accounted for.

SO **rank–nullity counts the input side**

`not m` the null space is not in the output space

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

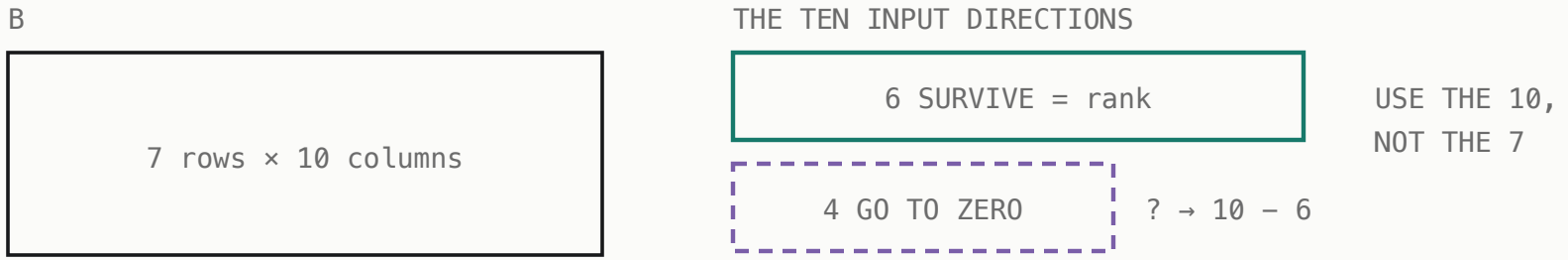
# Basis, Dimension, and Rank

EXAM QUESTION $B \in \mathbb{R}^{7\times 10}, \qquad \operatorname{rank}(B) = 6$

ask $\dim \operatorname{Col}(B)$, $\dim \operatorname{Null}(B)$, and the number of free variables

$$\dim \operatorname{Col}(B) = 6, \qquad \dim \operatorname{Null}(B) = 10 - 6 = 4, \qquad 4 \text{ free variables}$$

the trap use the 10, not the 7




$$\dim \operatorname{Null}(B) = n - r = 10 - 6 = 4.$$

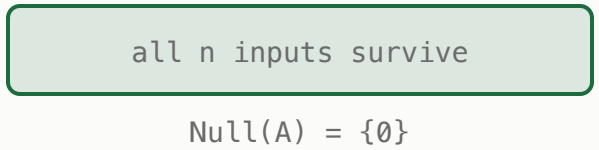
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

# Basis, Dimension, and Rank

**FULL COLUMN RANK**

$$r = n$$

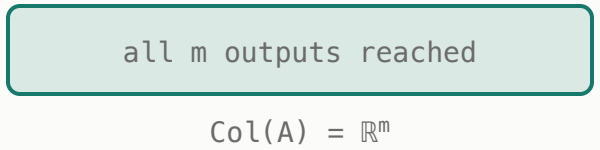
then $\text{Null}(A) = \{0\}$: no input direction is lost; needs $n \leq m$




**FULL ROW RANK**

$$r = m$$

then $\text{Col}(A) = \mathbb{R}^m$: every output is reachable; needs $m \leq n$




> One is about losing nothing. The other is about reaching everything.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

# Basis, Dimension, and Rank

WHERE WE ARE $\dim \mathrm{Col}(A) = r, \qquad \dim \mathrm{Null}(A) = n - r$



so the three questions become countable

$$\begin{aligned} \text{is } b \text{ reachable?} \quad & b \in \mathrm{Col}(A) \\ \text{is } x \text{ unique?} \quad & \mathrm{Null}(A) = \{0\} \\ \text{how many free directions?} \quad & n - r \end{aligned}$$

row two uniqueness is a question only once a solution exists

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

INTEGRATED WORKED EXAMPLE

# Basis, Dimension, and Rank

$F = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$. Find bases of $\mathrm{Col}(F)$ and $\mathrm{Null}(F)$, rank and nullity.

$$x_2 = s,\ x_4 = t: \quad x_1 = -2s - t,\ x_3 = t$$

reason Pivot columns 1 and 3 determine two variables.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# Basis, Dimension, and Rank

Continue; check the original map.

$$x_2 = s,\ x_4 = t: \quad x_1 = -2s - t,\ x_3 = t$$

reason Pivot columns 1 and 3 determine two variables.

$$\mathrm{Null}(F) = \mathrm{span}\{(-2, 1, 0, 0)^{\mathsf{T}}, (-1, 0, 1, 1)^{\mathsf{T}}\}$$

reason The two free-coordinate patterns prove independence.

$$\mathrm{Col}(F) = \mathbb{R}^2, \quad r = 2, \quad n - r = 2$$

reason Original columns 1 and 3 form the output basis; 2+2=4.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

INDEPENDENT TRANSFER

# Basis, Dimension, and Rank

$G = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$. Find rank, nullity and $\mathrm{Col}(G)$. Can it reach $(1, 2, 4)^{\mathsf{T}}$?

$$\operatorname{rank} G = 2, \quad \operatorname{nullity} G = 2 - 2 = 0$$

reason Two input coordinates, two independent columns.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# Basis, Dimension, and Rank

Continue; check the original map.

$$\operatorname{rank} G = 2, \quad \operatorname{nullity} G = 2 - 2 = 0$$

reason Two input coordinates, two independent columns.

$$Gx = (x_1, x_2, x_1 + x_2)^{\mathsf{T}}$$

reason Its image is the plane y3 = y1 + y2.

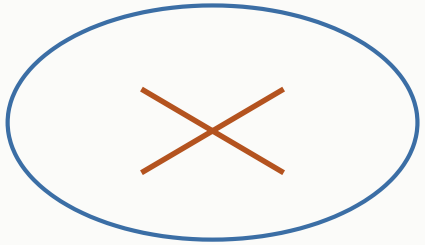
$$1 + 2 = 3 \neq 4$$

reason Full column rank gives uniqueness when reachable, not existence for every target.

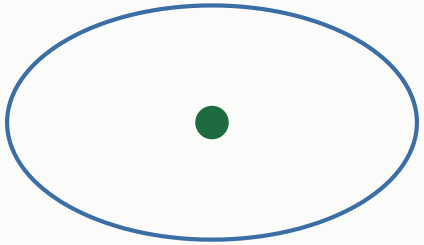
Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

# Linear Systems and Solution Sets

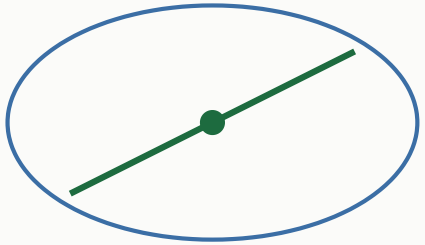
NONE



EXACTLY ONE



INFINITELY MANY



# Linear Systems and Solution Sets

RETURN TO $Ax = b$




`1 existence` does a solution exist?

`2 uniqueness` if it exists, is it the only one?

`3 structure` what are all of them?

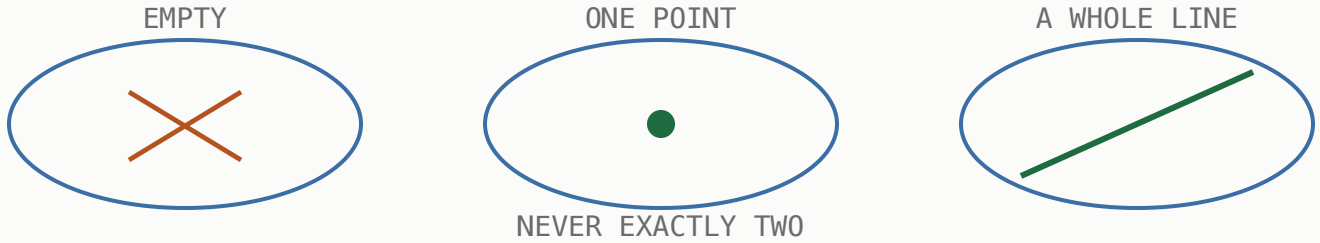
WITH $\mathrm{Col}(A)$ and $\mathrm{Null}(A)$

`nothing new` both spaces are already built and measured

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61; 3.11 null space, p. 59. Springer, 2024.

# Linear Systems and Solution Sets

**SOLUTION SET** $\mathcal{S}(A, b) = \{x \in \mathbb{R}^n : Ax = b\}$




`a solution` any x that produces the requested b

`consistent` $\mathcal{S}(A, b) \neq \emptyset$; otherwise inconsistent

**THREE POSSIBILITIES** empty, one point, infinitely many

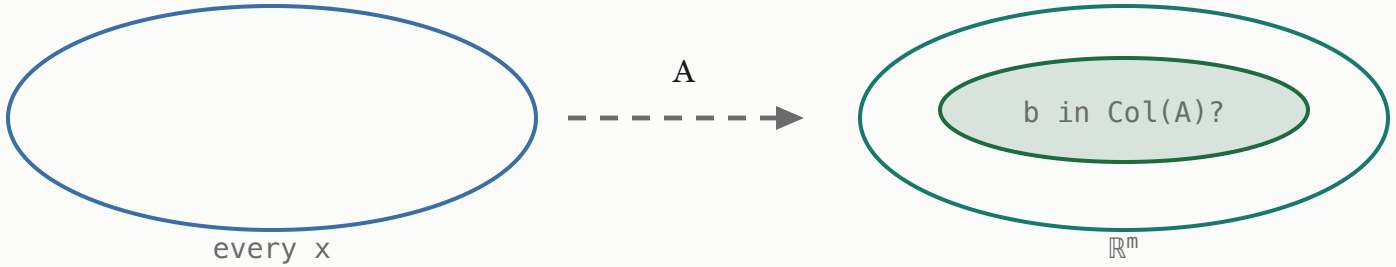
`never two` two solutions force infinitely many: a line, a plane, or higher

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Linear Systems and Solution Sets

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

RECALL $\mathrm{Col}(A) = \{Ax : x \in \mathbb{R}^n\}$




so b is reachable exactly when some x reaches it

EXISTENCE $\boxed{Ax = b \text{ is consistent} \iff b \in \mathrm{Col}(A)}$

one sentence the column space controls existence

**WORKED EXAMPLE**

$\mathrm{Col}(A)$ is $y_3 = 2y_1 + y_2$: $(3, 2, 8)$ passes since $2(3) + 2 = 8$; $(3, 2, 9)$ fails.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61. Springer, 2024.

# Linear Systems and Solution Sets

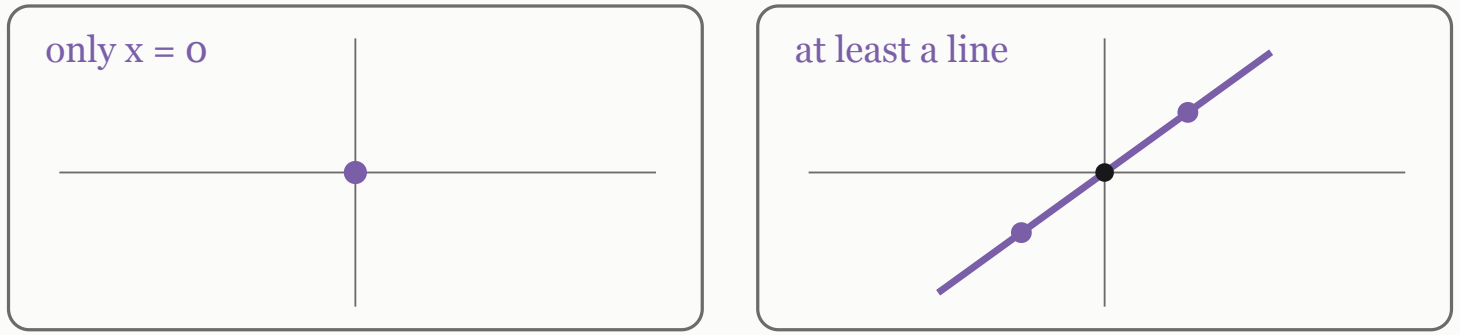
SUPPOSE $Ax_1 = b, \quad Ax_2 = b \implies A(x_1 - x_2) = 0 \implies x_1 - x_2 \in \text{Null}(A)$

so if $\text{Null}(A) = \{0\}$ then $x_1 = x_2$

UNIQUENESS $\boxed{\text{Null}(A) = \{0\} \implies \text{at most one solution}}$

careful at most one, not one

if consistent then it means exactly one

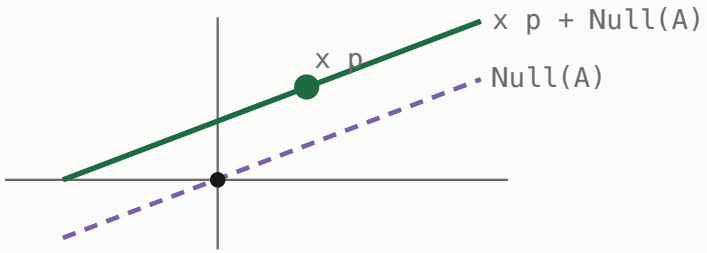



Either only the origin, or at least a whole line of erased inputs.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.15 injectivity, p. 60. Springer, 2024.

# Linear Systems and Solution Sets

PARTICULAR $Ax_p = b$ for one particular $x_p$




ONE SOLUTION, THEN EVERYTHING A IGNORES

`any one` it does not matter which solution you found

GENERAL $\mathcal{S}(A, b) = x_p + \text{Null}(A) = \{x_p + z : z \in \text{Null}(A)\}$

`recipe` find one solution, then add everything A ignores

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Linear Systems and Solution Sets

EACH ONE WORKS

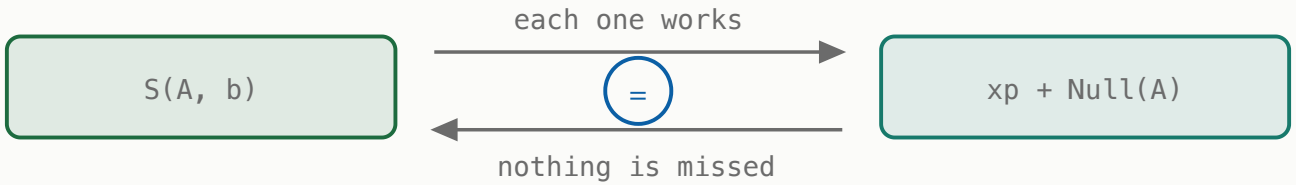
$$A(x_p + z) = b + 0 = b$$

uses linearity and $Az = 0$

NOTHING IS MISSED

$$Ax = b \Rightarrow A(x - x_p) = 0$$

so $x = x_p + z$ with $z = x - x_p \in \text{Null}(A)$




EQUALITY $\boxed{\mathcal{S}(A, b) = x_p + \text{Null}(A)}$

both ways one inclusion each, which is what an equality of sets needs

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Linear Systems and Solution Sets

AUGMENT

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 3 & 8 \end{array}\right] \xrightarrow{r_3 \leftarrow r_3 - 2r_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{array}\right] \xrightarrow{r_3 \leftarrow r_3 - r_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

step 1 $(2,1,3|8) - 2(1,0,1|3) = (0,1,1|2)$

step 2 $(0,1,1|2) - (0,1,1|2) = (0,0,0|0)$

row 3 $0 = 0$: consistent

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Linear Systems and Solution Sets

READ OFF $x_1 + x_3 = 3, \quad x_2 + x_3 = 2, \quad x_3 = t \implies x_1 = 3 - t,\ x_2 = 2 - t$

`free` no pivot in column 3: choose $x_3$

$t = 0$ one particular solution, $x_p = (3, 2, 0)^{\mathsf{T}}$

ALL OF THEM $\boxed{x = (3, 2, 0)^{\mathsf{T}} + t\,(-1, -1, 1)^{\mathsf{T}}, \qquad t \in \mathbb{R}}$

$t = 1$ $(2, 1, 1)^{\mathsf{T}}$: $2 + 1 = 3$, $1 + 1 = 2$, $4 + 1 + 3 = 8$

$b = (3, 2, 9)^{\mathsf{T}}$ same steps: $9 - 6 = 3$, $3 - 2 = 1$, so row 3 reads $0 = 1$: no solution

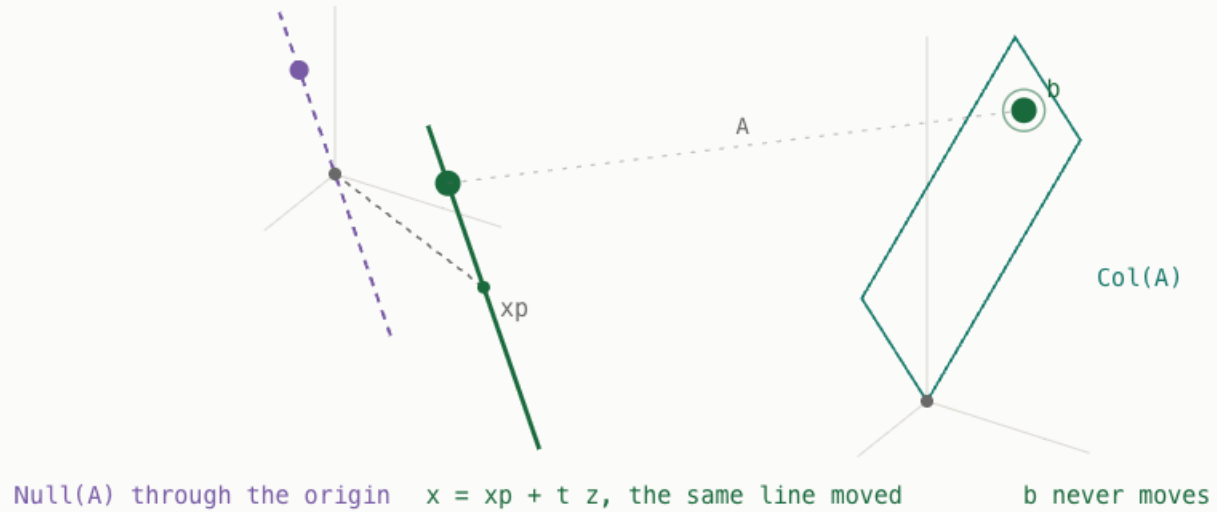
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

# Linear Systems and Solution Sets

AFFINE SET $\boxed{x_0 + S = \{x_0 + s : s \in S\} \text{ for a subspace } S}$

here $\mathcal{S}(A, b) = x_p + \text{Null}(A)$: the null space, translated

through 0? only if $b = 0$; then $\mathcal{S} = \text{Null}(A)$, a subspace




The null space through the origin, the same line moved to a particular solution, and an output that never moves.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.34 conditions for a subspace, p. 18. Springer, 2024.

# Linear Systems and Solution Sets

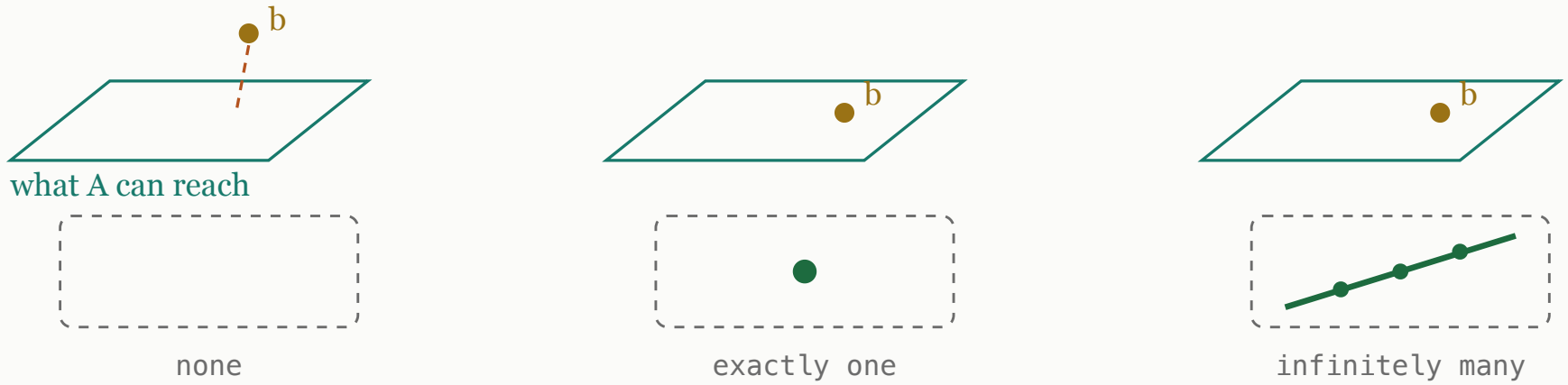
NOTATION $r = \operatorname{rank}(A)$

`unique` when consistent, means $r = n$

CASES

$$
\begin{aligned}
b \notin \operatorname{Col}(A) &: \text{ none} \\
b \in \operatorname{Col}(A),\ \operatorname{Null}(A) = \{0\} &: \text{ one} \\
b \in \operatorname{Col}(A),\ \dim \operatorname{Null}(A) > 0 &: \text{ infinitely many}
\end{aligned}
$$

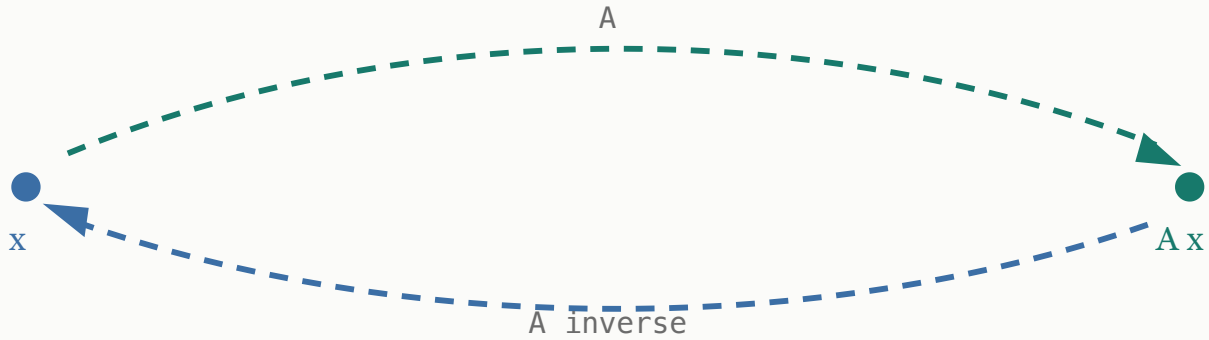
`free` $n - r$ directions




Top: the demand. Bottom: the solutions.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

# Invertibility and Singular Matrices



# Invertibility and Singular Matrices

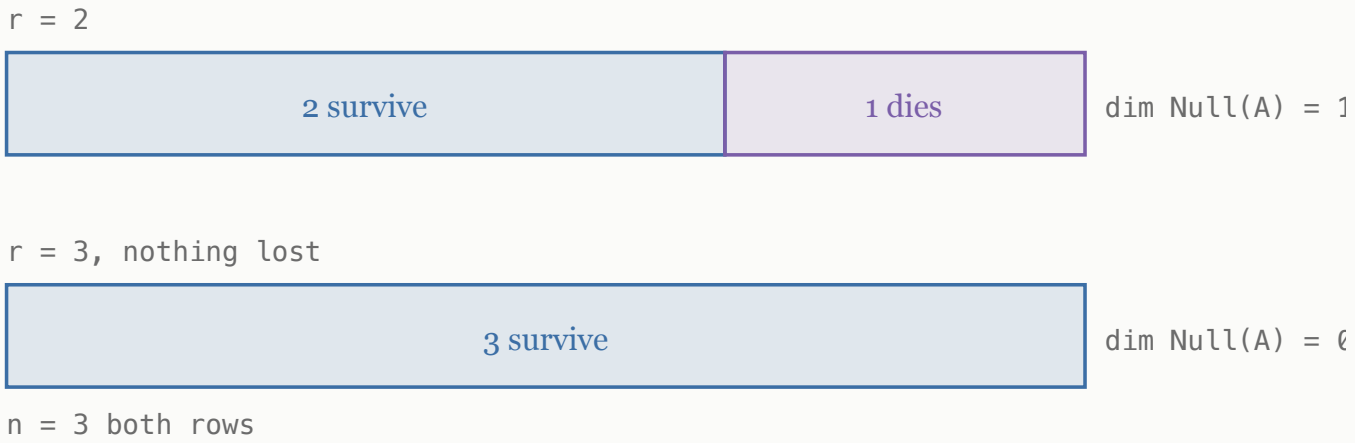
**DEMAND** $\text{Col}(A) = \mathbb{R}^m$ and $\text{Null}(A) = \{0\}$

`reach` $r = m$: nothing is missed

`no loss` $r = n$ by rank--nullity

`so` $\boxed{m = n \text{ and } r = n}$

`square` a conclusion, not a hypothesis




Both cases have three input directions to spend. Only the top spends one on the null space.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

# Invertibility and Singular Matrices

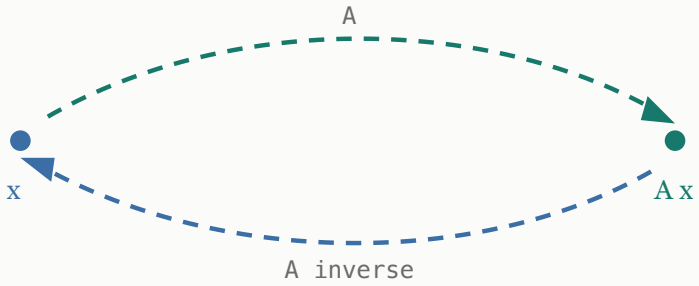
**INVERSE** $\boxed{AA^{-1} = A^{-1}A = I}$

**shapes** $A, A^{-1} \in \mathbb{R}^{n\times n}$ and $I = I_n$

**both sides** two-sided by definition; for square $A$ either half implies the other

**SINGULAR** $\boxed{A \text{ singular} \iff A \text{ has no inverse}}$

**nonsingular** the same as invertible




**WORKED EXAMPLE**

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.79 identity, p. 90; 3.80 invertible and the word singular, p. 91; 3.60 inverse is unique, p. 82. Springer, 2024.

# Invertibility and Singular Matrices

IN GENERAL $A$ invertible $\iff$ $A$ injective and surjective

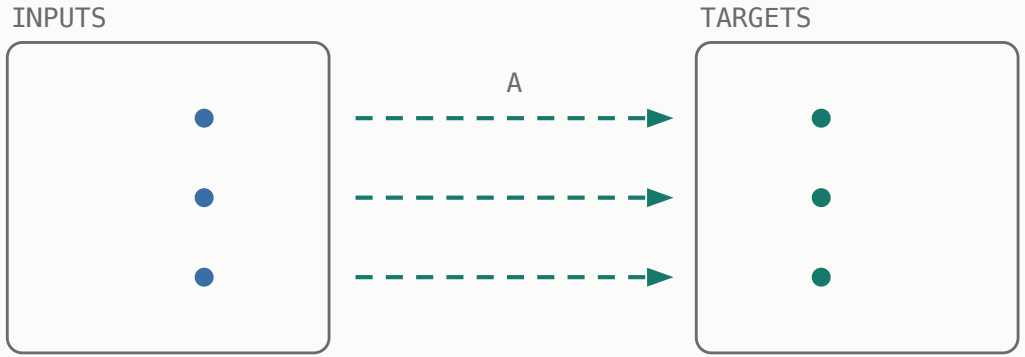
injective $Ax = Ay \Rightarrow x = y$

surjective $\mathrm{Col}(A) = \mathbb{R}^m$

IF A IS SQUARE $\boxed{\mathrm{Null}(A) = \{0\} \iff \mathrm{Col}(A) = \mathbb{R}^n \iff A \text{ invertible}}$

saving check one side, the other follows

why rank--nullity: $r = n$ makes both true at once




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.63 invertibility, p. 83; 3.65 injectivity equivalent to surjectivity, p. 84. Springer, 2024.

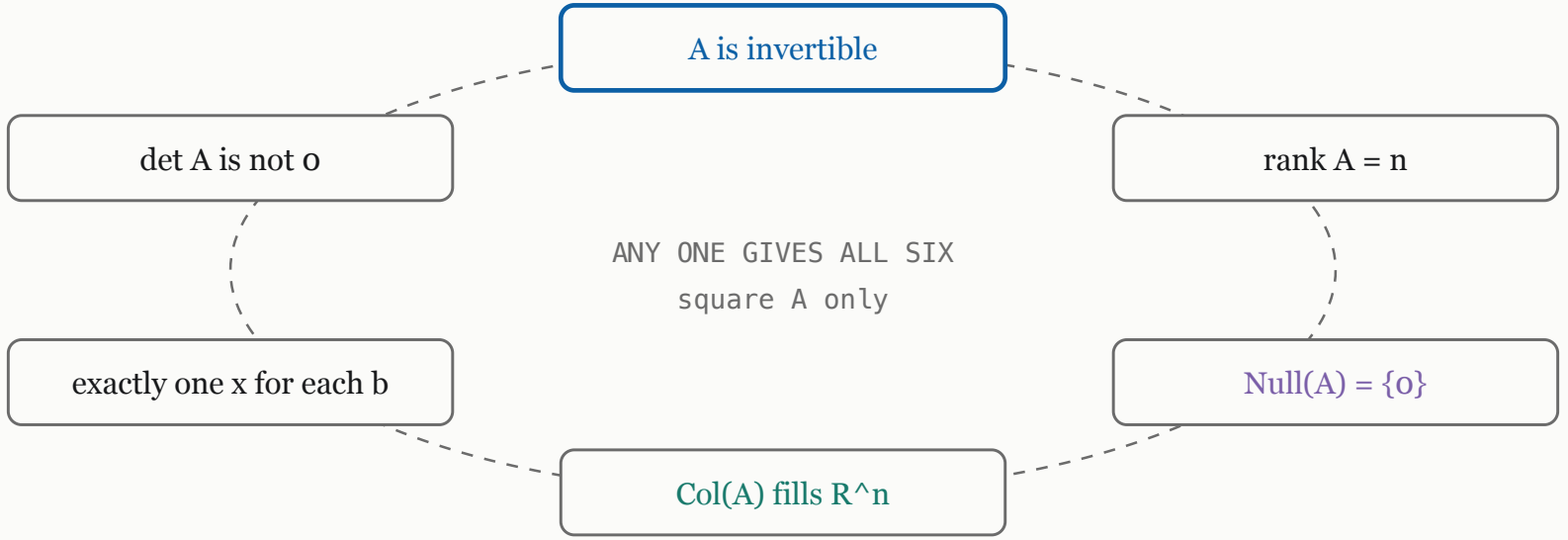
# Invertibility and Singular Matrices

SQUARE A

$$A \text{ invertible} \iff r = n \iff \text{Null}(A) = \{0\}$$
$$\iff \text{columns independent} \iff \text{Col}(A) = \mathbb{R}^n$$

and $\iff Ax = b$ has one solution for every $b$

det a preview; the determinant is a later lecture




Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.63 invertibility, p. 83; 3.65, p. 84; 3.21, p. 62. Springer, 2024.

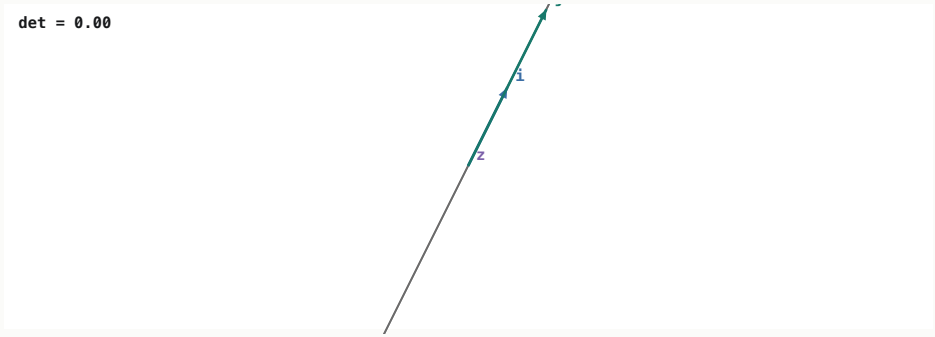
# Invertibility and Singular Matrices

SINGULAR $\boxed{\text{Null}(B) \neq \{0\} \iff B \text{ singular}}$

B square and rank deficient

**WORKED EXAMPLE**

$B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has $r = 1$, and $B(2, -1)^{\mathsf{T}} = 0$: a nonzero input lands on zero.




The unit square flattens; the area, and the inverse, are gone.

Sanderson, G. ***Essence of Linear Algebra***, ch. 7. 3Blue1Brown, 2016 · Axler, 4th ed., 3.80, p. 91.

# Invertibility and Singular Matrices

PREVIEW · LEAST SQUARES

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}, \qquad X^{\mathsf{T}}X = \begin{bmatrix} 1+4+9 & 2+8+18 \\ 2+8+18 & 4+16+36 \end{bmatrix} = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}$$

**the fit** $\min_{\beta} \|X\beta - y\|^2 \;\Rightarrow\; X^{\mathsf{T}}X\beta = X^{\mathsf{T}}y$

**entries** $(X^{\mathsf{T}}X)_{ij}$ is column $i$ dotted with column $j$; the off-diagonal pair agree

Goodfellow, I., Bengio, Y. and Courville, A. ***Deep Learning***, ch. 2. MIT Press, 2016.

# Invertibility and Singular Matrices

SINGULAR $$X\begin{bmatrix}2\\-1\end{bmatrix} = \begin{bmatrix}2-2\\4-4\\6-6\end{bmatrix} = 0 \implies X^{\mathsf{T}}X\begin{bmatrix}2\\-1\end{bmatrix} = 0$$

collinear column 2 is twice column 1: $(2,-1)^{\mathsf{T}} \in \mathrm{Null}(X)$

NULL SPACES $$\boxed{X^{\mathsf{T}}Xz = 0 \;\Rightarrow\; \|Xz\|^2 = z^{\mathsf{T}}X^{\mathsf{T}}Xz = 0 \;\Rightarrow\; Xz = 0}$$

so $\mathrm{Null}(X^{\mathsf{T}}X) = \mathrm{Null}(X)$

the fit $\beta + t(2,-1)^{\mathsf{T}}$ keeps $X\beta$: not identifiable

Goodfellow, I., Bengio, Y. and Courville, A. ***Deep Learning***, ch. 2. MIT Press, 2016.

# Invertibility and Singular Matrices

AT A SINGULARITY $J(q)$ loses rank $\Longrightarrow$ a tool direction is lost

**rank drop** $\mathrm{Col}(J)$ shrinks: some $\dot{x}$ has no $\dot{q}$

**nearby** joint rates grow without bound for tool directions along the lost one

NEXT $\mathrm{Col}(A),\ \mathrm{Null}(A),\ \mathrm{Col}(A^{\mathsf{T}}),\ \mathrm{Null}(A^{\mathsf{T}})$

**one matrix** four spaces, one rank




Lynch, K. M. and Park, F. C. ***Modern Robotics***, §5.3 singularity analysis, p. 191. Cambridge University Press, 2017 preprint.

INTEGRATED WORKED EXAMPLE

# Invertibility and Singular Matrices

$H = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $b = (5, 2)^{\mathsf{T}}$. Solve $Hx = b$, find $H^{-1}$, and connect uniqueness to rank.

$$x_2 = b_2, \quad x_1 = b_1 - 2b_2$$

reason Back-substitution works for an arbitrary target.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# Invertibility and Singular Matrices

Continue; check the original map.

$$x_2 = b_2, \quad x_1 = b_1 - 2b_2$$

reason Back-substitution works for an arbitrary target.

$$H^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \quad x = (1, 2)^{\mathsf{T}}$$

reason The inverse collects the coefficients of b.

$$HH^{-1} = H^{-1}H = I, \quad \mathrm{Null}(H) = \{0\},\ r = 2$$

reason A reversible map reaches every target exactly once.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

INDEPENDENT TRANSFER

# Invertibility and Singular Matrices

$K = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. Compare targets $(5, 10)^{\mathsf{T}}$ and $(5, 11)^{\mathsf{T}}$. Can an inverse exist?

$$b = (5, 10)^{\mathsf{T}}: \quad x = (5, 0)^{\mathsf{T}} + t(-2, 1)^{\mathsf{T}}$$

reason The second equation repeats twice the first.

Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

COMPLETE THE REASONING

# Invertibility and Singular Matrices

Continue; check the original map.

$$b = (5, 10)^{\mathsf{T}}: \quad x = (5, 0)^{\mathsf{T}} + t(-2, 1)^{\mathsf{T}}$$

reason The second equation repeats twice the first.

$$b = (5, 11)^{\mathsf{T}}: \quad 0 = 1$$

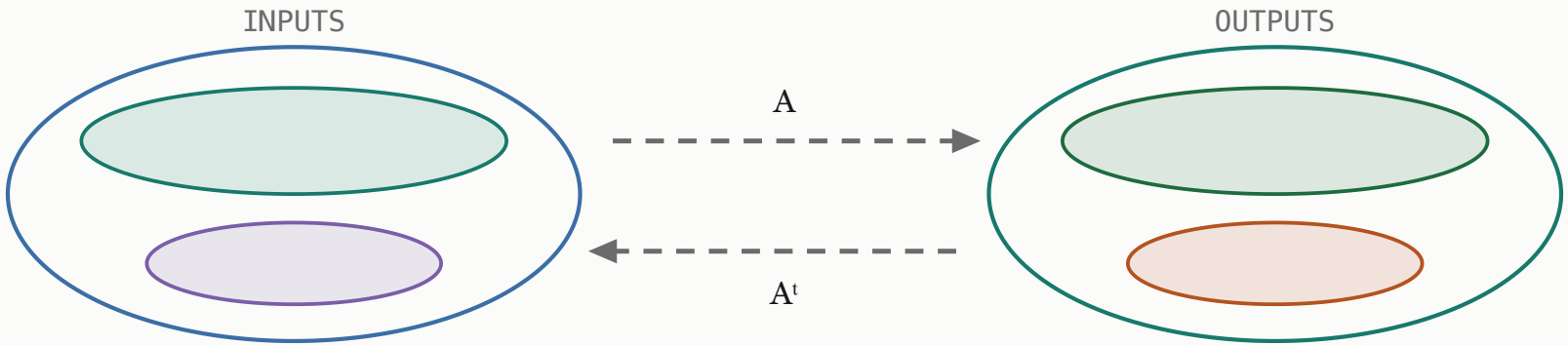
reason Subtract twice row 1 from row 2: inconsistent.

$$K(-2, 1)^{\mathsf{T}} = 0, \quad (-2, 1)^{\mathsf{T}} \neq 0$$

reason An inverse cannot recover two inputs that share an output.

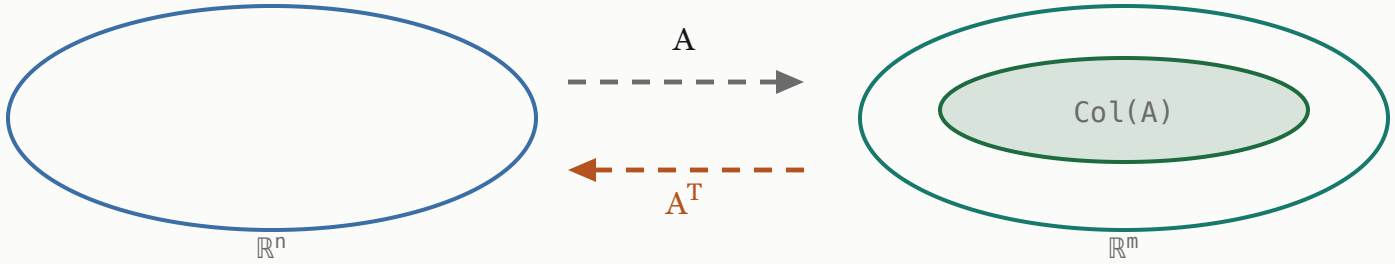
Course-authored example. Definitions: Axler, Linear Algebra Done Right, 4th ed., 2024; Strang, MIT 18.06.

# The Four Fundamental Subspaces



# The Four Fundamental Subspaces

SO FAR $A : \mathbb{R}^n \to \mathbb{R}^m$, $\quad \mathrm{Col}(A)$ and $\mathrm{Null}(A)$




but $A^\mathsf{T} : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map too

SO ASK **what spaces come from $A^\mathsf{T}$?**

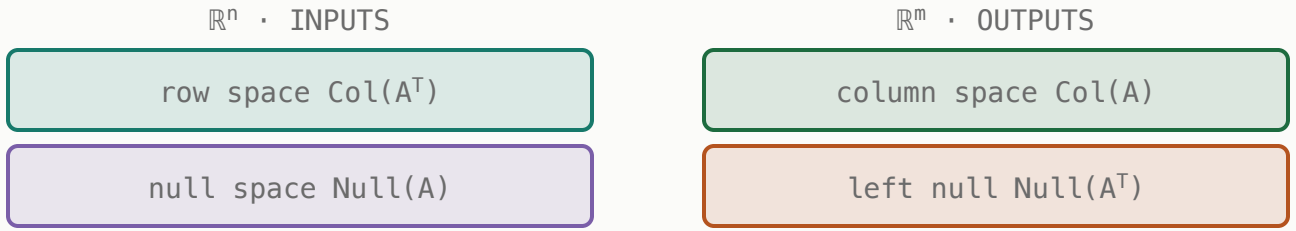
the answer $\mathrm{Col}(A^\mathsf{T})$ and $\mathrm{Null}(A^\mathsf{T})$: two more, for free

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.54 transpose, p. 77. Springer, 2024.

# The Four Fundamental Subspaces

NEW NAMES

$$\mathrm{Row}(A) = \mathrm{Col}(A^{\mathsf{T}}), \qquad \mathrm{Null}(A^{\mathsf{T}}) = \{y : A^{\mathsf{T}} y = 0\}$$




**left null** $A^{\mathsf{T}} y = 0$ is $y^{\mathsf{T}} A = 0^{\mathsf{T}}$: y multiplies from the left

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.54 transpose, p. 77; 3.11 null space, p. 59. Springer, 2024.

# The Four Fundamental Subspaces

OUTPUT SIDE

$$\mathrm{Col}(A) = \mathrm{span}\{(1,0,2)^{\mathsf{T}}, (0,1,1)^{\mathsf{T}}\}$$

INPUT SIDE

$$\mathrm{Row}(A) = \mathrm{span}\{(1,0,1)^{\mathsf{T}}, (0,1,1)^{\mathsf{T}}\}$$

$\mathrm{Null}(A^{\mathsf{T}}) = \mathrm{span}\{(-2,-1,1)^{\mathsf{T}}\}$

$\mathrm{Null}(A) = \mathrm{span}\{(-1,-1,1)^{\mathsf{T}}\}$

```
row 1   1 0 1
row 2   0 1 1      2r1 + r2 = r3
row 3   2 1 3      so −2r1 − r2 + r3 = 0
```

so $\boxed{y = (-2,-1,1)^{\mathsf{T}}, \qquad A^{\mathsf{T}}y = 0}$

`read it` the row weights

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.57 column rank equals row rank, p. 78. Springer, 2024.

# The Four Fundamental Subspaces

RANK EQUALITY $$\boxed{\operatorname{rank}(A) = \operatorname{rank}(A^{\mathsf{T}}) = r}$$




the output side: r = 2, m − r = 1

so column rank equals row rank

ALL FOUR $$\boxed{\begin{aligned} \dim \operatorname{Col}(A) &= r & \dim \operatorname{Null}(A) &= n - r \\ \dim \operatorname{Col}(A^{\mathsf{T}}) &= r & \dim \operatorname{Null}(A^{\mathsf{T}}) &= m - r \end{aligned}}$$

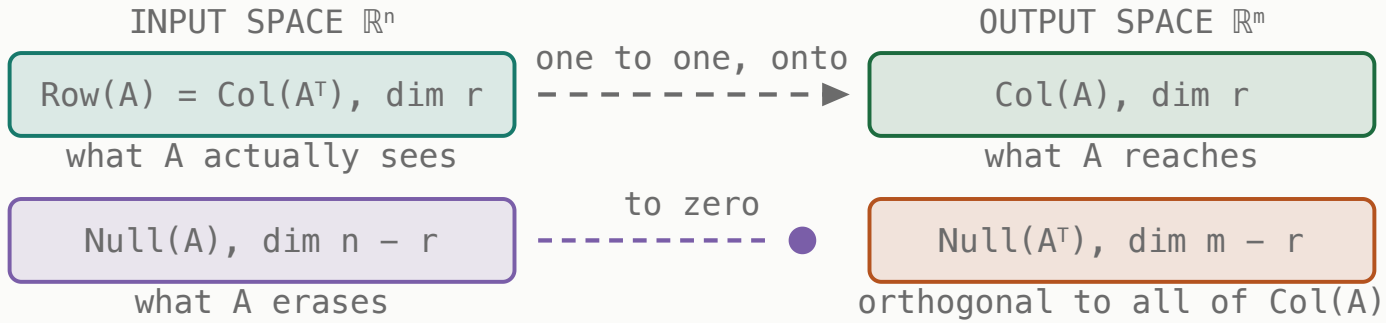
two sums $r + (n - r) = n$ and $r + (m - r) = m$

**WORKED EXAMPLE**

Our $A$: $r = 2$, so $2 + 1 = 3$ columns and $2 + 1 = 3$ rows.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.57 column rank equals row rank, p. 78; 3.21, p. 62. Springer, 2024.

# The Four Fundamental Subspaces




**LECTURE 1 IN ONE LINE** $\dim = r,\ n-r,\ r,\ m-r$

`later` the two pairs form orthogonal-complement decompositions

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21, p. 62; 3.57, p. 78. Springer, 2024.

INTEGRATED WHITEBOARD PROBLEM

# Linear Systems and Four Subspaces

Given $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix}$, $b = (4, 1, 5)^{\mathsf{T}}$. Find reachability, all solutions and four-space dimensions.

$$r_3 - r_1 - r_2 = 0 \;\Rightarrow\; b_3 = b_1 + b_2, \quad r = 2$$

`r` rank; two independent rows

$$x = (2, 1, 0)^{\mathsf{T}} + t(2, -1, 1)^{\mathsf{T}}$$

`t` any real; dimensions: $2, 1, 2, 1$

YOUR TURN · NEW MATRIX

# Linear Systems and Four Subspaces

$D = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$, $b = (3, 1)^{\mathsf{T}}$. Find all solutions. Is every target reachable? Is its input unique?

$$x_3 = t, \quad x_1 = 3 - 2t, \quad x_2 = 1 + t$$

`t` the free third coordinate

$$x = (3, 1, 0)^{\mathsf{T}} + t(-2, 1, 1)^{\mathsf{T}}$$

`rank 2` every target reached; nullity 1: inputs not unique

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# Courses, books and lectures


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